

ULTIMATE CYBERSECURITY MASTER BLUEPRINT 2026-2027

Volume 5 — Resource Directory

169 entries, coverage matrix, gap analysis, playlist policy and verification register

Edition: Volume 5
Resource verification date: 2026-10-27
Plan window: 2026-10-27 to 2027-10-26 (shift dates if you start later)
System: 365 days, 12 academic months, 52 weeks + 1 graduation day
Profile: BS CS student in Pakistan; free/high-value resources; safe authorized labs

Academic-style self-study manual. No employment, salary, visa, certification or top-1% guarantee.

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Detailed resource cards (R001-R169) **21**

Resource directory and verification

169	162	7	0
verified registry entries	HTTP 200 responses	automated-blocker responses	404 links remaining

Full directory follows. Each card includes exact title, direct URL, type, topic/subtopic coverage, exact path, difficulty, time, access, prerequisites, selection rationale, post-study practice, backup, limitations, classification, age status and verification result.

Volume 5 — Resource directory, coverage matrix and verification register

Verification date: 2026-10-27. Every URL in the registry was checked with an automated HTTP request from this workspace. A `200` means the endpoint responded; it does not guarantee that a specific paid/free feature remains available in every country. A `403`, `429`, `202` or TLS issue is explicitly labelled as an automated-checker limitation and has a backup resource. No link was assumed to work simply because it looked plausible.

Selection principles

1. **Primary source bias.** Official documentation, university courses, standards and project-maintained labs outrank popularity.
2. **One primary per objective.** Supplementary material is used for alternate explanation and coverage gaps, not endless course collecting.
3. **Free first.** Paid resources are labelled and always paired with a free alternative.
4. **Current plus old-but-gold.** Older material is used only when the model/concept remains durable; dated commands and tool UIs must be checked against current docs.
5. **No invented timestamps.** For video, the PDF names the course, playlist, module or objective code. Use the chapters actually visible on the video page.
6. **Legal practice.** Lab platforms and vulnerable apps are used only in their authorized form.

Classification

- **Mandatory** — core to this curriculum.
- **Recommended** — fills important gaps or provides a better alternate explanation.
- **Optional enrichment** — useful when time/hardware allows.
- **Advanced reference** — deeper source for Year 2+; not a Year 1 mastery target.
- **Paid/alternative** — paid resource with a free replacement.

Coverage matrix and gaps

The matrix uses module letters A-K defined in Volume 1. ● = primary coverage; ○ = supporting/partial; blank = not intended. Exact supplementary links are in the resource cards.

Domain	Primary	Supplementary	Lab/practice	Authoritative reference	Assessment/ challenge	Main gap and fix
A CS/IT foundations	CS50x 2026 (F01)	OSTEP (F06), Nand2Tetris (F07)	VM setup (F12), Linux Journey (L01)	Python/Git/ MDN docs	CS50 problem sets	Hardware timing is advanced; F07 optional projects supply it
B Networking	Jeremy's IT Lab (N01)	Professor Messer	Packet Tracer (N15),	RFC 9293/791/820	Subnetting Practice (N16),	IPv6/security details need RFCs; N12 fills header gaps

Domain	Primary	Supplementary	Lab/practice	Authoritative reference	Assessment/ challenge	Main gap and fix
		N10-009 (N04)	Wireshark samples (N07)	0, IANA ports (N14)	Wireshark reports	
C Linux and Windows	Linux Journey (L01) + Microsoft docs (L07-L14)	BashGuide (L03), man7 (L04)	OverTheWire Bandit (L02), pwn.college (L15)	systemd/man-pages/ Windows Event Log docs	Bandit, event-log investigation	Full AD deployment needs 16 GB+; diagram-based alternative provided
D Programming	Python Tutorial (F08) + CS50 (F01)	Automate the Boring Stuff (P03), py4e (P01)	RegexOne (P04), pytest (P05), SQLite (P07)	Python/ Requests/Git docs	Month 4 log parser tool	Java/.NET secure coding not taught; OWASP cheat sheets are language-neutral
E Core security	CS50 Cyber (F04) + Professor Messer (C01)	NIST CSF/NIST SP docs	ISC2 free flashcards, Crypto 101 exercises	NIST CSF, NIST SP 800-63B, RFC 8446	Month 5 risk/ crypto practical	Formal proofs deferred to Stanford/graduate crypto resources
F Web/ API	PortSwigger Academy (W01/W02)	OWASP Top 10/API Top 10/ ASVS	PortSwigger labs, Juice Shop/WebGoat	MDN HTTP, OWASP Cheat Sheets	Apprentice labs + secure API project	Advanced request smuggling/GraphQL are Year 2+; W03 expert labs postponed
G Ethical testing	NIST SP 800-115 (E01)	OWASP WSTG (W06)	Local VMs, HTB Academy free modules	FIRST CVSS, CVE/NVD, HackerOne guidance	PRJ-10 scoped report	Real-world commercial scoping requires supervised experience
H Blue team	Microsoft Sentinel/KQL/ Sigma (B01/ B02/B03)	MITRE ATT&CK, NIST 800-61r3	CyberDefenders, BTLO, EVTX samples	Microsoft/ Sigma/Elastic official docs	PRJ-11/12	Paid SIEM ingestion avoided; synthetic/local logs used instead
I Cloud/ Infra	AWS Security Pillar + Docker/K8s docs	Microsoft Azure, OWASP K8s Top 10	Docker/kind/ minikube, Trivy, Checkov	AWS/Azure/ K8s/GitHub official docs	PRJ-13/14	No expensive managed cluster; local control-plane emulation has limits
J GRC	NIST CSF 2.0 (C06)	CIS Controls, ISO overview, GDPR text	Tabletop risk/IR exercises	NIST SP 800-30/34/53, EUR-Lex	M5/M12 risk register	Full ISO standard is paid; legal applicability needs a qualified adviser
K Advanced	pwn.college, OST2, MIT 6.858	Security Engineering, USENIX, Project Zero	pwn.college dojos, Malware Unicorn, Nightmare	Papers, standards, advisories	Year 2+ reproduction	Years of C/assembly/ systems prerequisites; not mastered in Year 1

Gap analysis: exact supplements

Missing/weak subtopic	Why primary course omits it	Exact supplement
CPU datapath and sequential logic	CS50 is broad and abstract	Nand2Tetris Part I Projects 1-3 (F07)
OS process/memory mechanisms	Intro courses emphasize usage	OSTEP Chapters 4-20 and 53-57 (F06)
IPv6 extension headers and modern TCP state machine	CCNA course is exam-scoped	RFC 8200 (N12) and RFC 9293 (N10)
Linux kernel/user API exact semantics	Linux Journey is beginner text	man7 pages (L04) and systemd.unit (L05)
Windows event schema and Sysmon	Intro admin courses focus on UI	Windows Event Log (L09), Sysmon (L11), EVTX samples (B05)

Missing/weak subtopic	Why primary course omits it	Exact supplement
Secure coding verification	PortSwigger is exploit-first	OWASP ASVS (C15) and Cheat Sheets (W17)
API object-level authorization	General OWASP Top 10 is broad	OWASP API Top 10 (C16) and local API project
Detection rule portability	SIEM docs are vendor-specific	SigmaHQ (B03/B04)
Cloud control catalog	AWS docs are service-specific	CSA CCM (CL16), NIST CSF (C06)
Privacy legal distinctions	Frameworks are voluntary	GDPR official text (C24) and NIST Privacy Framework (C25)
Research paper critique	Cert courses use summaries	MIT 6.858 (A06) and USENIX Security (A12)
Low-level RE prerequisites	Tool tutorials assume assembly	OST2 Arch1001 (A04) and pwn.college Linux Luminarium (A01)
AI application security	Classical web courses predate LLM apps	OWASP LLM Top 10 (A15) and NIST AI RMF (A16)

Do not waste time

- **Do not complete three beginner courses for the same topic.** Use one primary and one supplementary.
- **Do not watch entire playlists sequentially when a single objective is weak.** Use the course index by objective code.
- **Do not collect tools.** Master Wireshark, Nmap, Burp, Python, Git, Docker, KQL/Sigma and one cloud model deeply.
- **Do not buy exam vouchers for motivation.** Buy only after current objectives, free practice and a concrete job/education reason.
- **Do not buy SANS/GIAC, OSCP, BTL1 or HTB subscriptions in Month 1.** Their value depends on prior foundations and supervised practice.
- **Do not copy write-ups.** Struggle, hint, docs, then solution; always re-solve with a variation.
- **Do not deploy vulnerable apps publicly.** Use loopback/host-only/internal networks and snapshots.
- **Do not confuse framework compliance with security.** A control matrix is not evidence that a control works.

Lesser-known but credible resources

Resource	What it teaches better than a generic alternative	Prerequisite	Safe use
OSTEP (F06)	OS mechanisms, not just commands; security chapters added	C/Unix basics	Selected chapters, homework questions
OpenSecurityTraining2 (A03/A04)	x86-64 assembly from a systems-security viewpoint	C and binary arithmetic	Year 2+, disposable VM
pwn.college (A01/L15)	University-grade Linux, web, memory and exploitation challenge ladder	Linux/C/Python	Free dojo, browser workspace, no external targets
Practical Networking Subnetting Mastery (N17)	Visual subnetting method, avoids rote tricks	Binary/IP	Month 2 daily practice
Greg's Wiki BashGuide (L03)	Correct quoting and word-splitting model	Basic shell	Audit scripts only on own files
EVTX-ATTACK-SAMPLES (B05)	Realistic synthetic Windows telemetry for detection tests	Windows logs	Data only; isolated VM; do not execute binaries
SigmaHQ (B03/B04)	Vendor-neutral detection-as-code rules	Logs/ATT&CK	

Resource	What it teaches better than a generic alternative	Prerequisite	Safe use
			Write rules, test on synthetic events
Malware Traffic Analysis (B08)	Realistic PCAP exercise archive	Wireshark/TCP/IP	Never execute samples; use disposable VM
Security Engineering 3e (A08)	Technical, human and economic failure modes together	Broad security basics	One chapter/month in Year 2+
Nightmare / RPISEC MBE (A09/A10)	Progressive binary-exploitation challenge structure	C, assembly, GDB, Linux	Year 2-3 only, isolated VMs, modern mitigations checked
Project Zero / USENIX / CISA KEV	Root-cause research, peer-reviewed work, exploitation evidence	Advanced domain knowledge	Read and reproduce safe measurements only
SLSA / OpenSSF Scorecard / Semgrep	Concrete software supply-chain controls, not vague DevSecOps slogans	Git/CI	Use on sanitized public repos
Kubernetes Goat / OWASP K8s Top 10	Hands-on cluster misconfigurations without public cloud	Docker/K8s basics	Local disposable cluster, 16 GB+ RAM

YouTube publication-date policy

- Check the individual video's upload date on YouTube before citing it.
- If a course index lists an objective code/version, that version controls more than the upload date.
- **Old but gold** means core models remain valid (TCP, x86, crypto primitives, OS mechanisms), not that all commands/UI are current.
- Security+ SY0-701 videos are useful for concepts until V8/SY0-801 launches; do not use them as the sole exam prep after the version switch.
- Jeremy's IT Lab playlist originated in 2020 and has been augmented; verify current Cisco exam topics before certification.
- MIT 6.858 (2014) and Malware Unicorn RE101 (2019) are pedagogically strong but require updated tools and VM images.

Resource verification register — all 169 resources

Status is from the automated HTTP check on 2026-10-27. A 200 means reachable, not that every regional free/paid feature is guaranteed. 403/429/202 are automated-checker limits, not evidence that the official link is fake. No 404 links remain in the required registry; backups are provided.

ID	Resource	URL	Topic	Verification status	Access	Date checked	Known limitation	Backup
F01	CS50x 2026 OpenCourseWare	https://cs50.harvard.edu/x/	A,D	Verified HTTP 200; endpoint reachable	Free; paid edX verified certificate optional	2026-10-27	Annual edition changes; use 2026 or current site	F02 — CS50x Lectures 2026 (official YouTube playlist)
F02	CS50x Lectures 2026 (official YouTube playlist)	https://www.youtube.com/playlist?list=PLhQjrBD2T380hlTqAU8HfvVepCcjCqTg6	A,D	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Playlist titles can be renumbered annually; match the official week page first	F01 — CS50x 2026 OpenCourseWare
F03	CS50x Sections 2026 (official YouTube playlist)	https://www.youtube.com/playlist?list=PLhQjrBD2T3839iqxqUdw7NWOCboh00vsV	A,D	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Older section videos may use previous tool versions	F02 — CS50x Lectures 2026 (official YouTube playlist)
F04	CS50's Introduction to Cybersecurity	https://cs50.harvard.edu/cybersecurity/	E	Verified HTTP 200; endpoint reachable	Free OCW; paid edX verified certificate optional	2026-10-27	High level; does not teach SOC tooling, exploitation, or secure coding in depth	F05 — CS50's Introduction to Cybersecurity (official YouTube playlist)
F05	CS50's Introduction to Cybersecurity (official YouTube playlist)	https://www.youtube.com/playlist?list=PLhQjrBD2T383Cqo5I1oRrbC1EKRAKGKUE	E	Verified HTTP 200; endpoint reachable	Free	2026-10-27	2023 publication date; core concepts remain valid, current product interfaces may differ	F04 — CS50's Introduction to Cybersecurity
F06	Operating Systems: Three Easy Pieces (OSTEP)	https://pages.cs.wisc.edu/~remzi/OSTEP/	A,C	Verified HTTP 200; endpoint reachable	Free online chapters; paid print optional	2026-10-27	Some examples assume Unix/C familiarity; do not attempt every chapter in Month 1	F01 — CS50x 2026 OpenCourseWare
F07	Nand2Tetris / The Elements of Computing Systems	https://www.nand2tetris.org/	A	Verified HTTP 200; endpoint reachable	Free software and course materials; book optional paid	2026-10-27	Large time commitment; use selected projects only	F06 — Operating Systems: Three Easy Pieces (OSTEP)
F08	Python 3 Official Tutorial	https://docs.python.org/3/tutorial/	A,D	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Reference style can be terse for absolute beginners	P02 — Learn Python - Full Course for Beginners (freeCodeCamp)
F09	Pro Git, 2nd Edition	https://git-scm.com/book/en/v2	A,D	Verified HTTP 200; endpoint reachable	Free	2026-10-27	GUI screenshots may differ from local tools	P01 — Python for Everybody
F10	MDN HTTP Overview	https://developer.mozilla.org/en-US/docs/Web/HTTP/Guides/Overview	A,B,F	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Some pages move as MDN reorganizes; use site search if a URL changes	W04 — MDN HTTP documentation
F11	SQLBolt	https://sqlbolt.com/	A,D	Verified HTTP 200; endpoint reachable	Free	2026-10-27	SQLite dialect differs slightly from	

ID	Resource	URL	Topic	Verification status	Access	Date checked	Known limitation	Backup
							PostgreSQL or SQL Server	P08 — OWASP Secure Coding Practices Quick Reference Guide
F12	Oracle VirtualBox User Manual	https://www.virtualbox.org/manual/	A,C	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Version-specific menus may differ; use the manual matching your installed version	L07 — Microsoft Windows Server documentation
F13	Ubuntu Desktop Guide	https://help.ubuntu.com/	A,C	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Release-specific UI changes	L01 — Linux Journey
F14	Microsoft Learn Windows fundamentals	https://learn.microsoft.com/en-us/training/modules/explore-windows-client/	A,C	Verified HTTP 200; endpoint reachable	Free learning content; exams separate	2026-10-27	Product names and UI change; rely on current Microsoft Learn pages	L08 — PowerShell documentation
N01	Jeremy's IT Lab Free CCNA 200-301 Complete Course	https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ	B	Verified HTTP 200; endpoint reachable	Free on YouTube; optional paid course extras	2026-10-27	Playlist originated in 2020 and was augmented for v1.1; verify current Cisco blueprint before certification study	N02 — Jeremy's IT Lab CCNA Course Index and Files
N02	Jeremy's IT Lab CCNA Course Index and Files	https://courses.jeremysitlab.com/p/ccna	B	Verified HTTP 200; endpoint reachable	Free YouTube course; paid ad-free edition has bonus quizzes	2026-10-27	Paid edition is optional; free files are linked separately	N03 — Jeremy's IT Lab CCNA Course Files
N03	Jeremy's IT Lab CCNA Course Files	https://jitl.jp/ccna-files	B	Verified HTTP 200; endpoint reachable	Free	2026-10-27	External host may change; use N02 video description if unavailable	N02 — Jeremy's IT Lab CCNA Course Index and Files
N04	Professor Messer N10-009 Network+ Training Course	https://www.youtube.com/playlist?list=PLG49S3nxzAnl_tQe3kvnmeMid0mjF8Le8	B	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Certification-oriented, not a complete packet-analysis course	N05 — Professor Messer N10-009 Course Index
N05	Professor Messer N10-009 Course Index	https://www.professormesser.com/network-plus/n10-009/n10-009-video/n10-009-training-course/	B	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Older N10-008 course also exists; ensure N10-009	N04 — Professor Messer N10-009 Network+ Training Course
N06	Wireshark User's Guide	https://www.wireshark.org/docs/wsug_html_chunked/	B,H	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Version-specific UI; match your installed Wireshark version	N07 — Wireshark SampleCaptures
N07	Wireshark SampleCaptures	https://wiki.wireshark.org/SampleCaptures	B,H	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Some linked external captures disappear; download from stable Wireshark-hosted mirrors where possible	N06 — Wireshark User's Guide
N08	tcpdump manual page	https://www.tcpdump.org/manpages/tcpdump.1.html	B,H	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Root privileges may be needed; do not capture networks you do not own	N06 — Wireshark User's Guide

ID	Resource	URL	Topic	Verification status	Access	Date checked	Known limitation	Backup
N09	Nmap Reference Guide	https://nmap.org/book/man.html	B,G	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Many options are not appropriate for beginners or public networks	N10 — RFC 9293: Transmission Control Protocol
N10	RFC 9293: Transmission Control Protocol	https://www.rfc-editor.org/rfc/rfc9293	B	Verified HTTP 200; endpoint reachable	Free	2026-10-27	RFC-level prose is dense; use after guided lessons	B10 — Suricata documentation
N11	RFC 791: Internet Protocol v4	https://www.rfc-editor.org/rfc/rfc791	B	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Historic RFC; use current clarifications for security specifics	N10 — RFC 9293: Transmission Control Protocol
N12	RFC 8200: Internet Protocol v6	https://www.rfc-editor.org/rfc/rfc8200	B	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Updated by later RFCs; use as baseline	N10 — RFC 9293: Transmission Control Protocol
N13	Cloudflare Learning Center: What is DNS?	https://www.cloudflare.com/learning/dns/what-is-dns/	B	Automated checker blocked (HTTP 403); official/canonical URL; use backup if browser access fails	Free	2026-10-27	Vendor educational content; cross-check record behavior in packet captures	B10 — Suricata documentation
N14	IANA Service Name and Transport Protocol Port Number Registry	https://www.iana.org/assignments/service-names-port-numbers/service-names-port-numbers.xhtml	B,G	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Registration is not proof a service uses that port in a particular environment	N10 — RFC 9293: Transmission Control Protocol
N15	Cisco Packet Tracer course page	https://www.netacad.com/courses/packet-tracer	B	Verified HTTP 200; endpoint reachable	Free with NetAcad account availability varies by region	2026-10-27	Registration and local availability may vary	N03 — Jeremy's IT Lab CCNA Course Files
N16	Subnetting Practice	https://subnettingpractice.com/	B	Verified HTTP 200; endpoint reachable	Free tier; optional ads/ premium features may appear	2026-10-27	Not a conceptual textbook; use only after the lesson	N17 — Practical Networking Subnetting Mastery
N17	Practical Networking Subnetting Mastery	https://www.practicalnetworking.net/stand-alone/subnetting-mastery/	B	Verified HTTP 200; endpoint reachable	Free articles/videos; optional paid resources	2026-10-27	Course packaging and optional product prices may change	N16 — Subnetting Practice
L01	Linux Journey	https://linuxjourney.com/	C	Automated checker blocked (HTTP 403); official/canonical URL; use backup if browser access fails	Free	2026-10-27	Does not replace distribution documentation or systemd internals	L02 — OverTheWire Bandit
L02	OverTheWire Bandit	https://overthewire.org/wargames/bandit/	C	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Occasional service availability; some late levels rely on Unix tricks	L01 — Linux Journey
L03	Greg's Wiki BashGuide	https://mywiki.wooledge.org/BashGuide	C,D	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Wiki can be terse; use as a reference while coding	P04 — RegexOne
L04	Linux man-pages project	https://man7.org/linux/man-pages/	C	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Distribution versions may differ; compare with local man pages	L01 — Linux Journey
L05	systemd.unit manual		C		Free	2026-10-27		

ID	Resource	URL	Topic	Verification status	Access	Date checked	Known limitation	Backup
		https://www.freedesktop.org/software/systemd/man/latest/systemd.unit.html		Verified HTTP 200; endpoint reachable			Not all distributions use identical defaults	L04 — Linux man-pages project
L06	The Linux Foundation Introduction to Linux	https://www.edx.org/learn/linux/the-linux-foundation-introduction-to-linux	C	Verified HTTP 200; endpoint reachable	Free audit may be time-limited; verified certificate paid	2026-10-27	edX audit terms and course versions vary	L01 — Linux Journey
L07	Microsoft Windows Server documentation	https://learn.microsoft.com/en-us/windows-server/	C	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Broad documentation; use the daily plan's exact pages	L12 — Active Directory Domain Services overview
L08	PowerShell documentation	https://learn.microsoft.com/en-us/powershell/	C,D	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Windows PowerShell 5.1 and PowerShell 7 differ; check edition	L03 — Greg's Wiki BashGuide
L09	Windows Event Log documentation	https://learn.microsoft.com/en-us/windows/win32/wes/windows-event-log	C,H	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Some APIs are developer-oriented; use Event Viewer for first practice	L11 — Sysmon documentation and download
L10	Microsoft Sysinternals	https://learn.microsoft.com/en-us/sysinternals/	C,H	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Tools change; use current downloads and hashes	L08 — PowerShell documentation
L11	Sysmon documentation and download	https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon	H	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Do not deploy on production systems without organizational approval	L09 — Windows Event Log documentation
L12	Active Directory Domain Services overview	https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/get-started/virtual-dc/active-directory-domain-services-overview	C,E	Verified HTTP 200; endpoint reachable	Free	2026-10-27	UI and server version differ across environments	L07 — Microsoft Windows Server documentation
L13	Windows security baselines	https://learn.microsoft.com/en-us/windows/security/operating-system-security/device-management/windows-security-configuration-framework/windows-security-baselines	C,E	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Do not apply a baseline blindly to production or your primary computer	L14 — CIS Benchmarks
L14	CIS Benchmarks	https://www.cisecurity.org/cis-benchmarks	C,E,I,J	Verified HTTP 200; endpoint reachable	Free PDFs may require registration; some tools paid	2026-10-27	Registration and profile versions; not all benchmarks fit student hardware	L13 — Windows security baselines
L15	pwn.college Dojo index	https://pwn.college/dojos	A,C,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Dojo names and course instances change; use	L01 — Linux Journey

ID	Resource	URL	Topic	Verification status	Access	Date checked	Known limitation	Backup
							current Core Material list	
P01	Python for Everybody	https://www.py4e.com/	D	Verified HTTP 200; endpoint reachable	Free site; Coursera specialization freemium	2026-10-27	Some videos are older; Python 3 syntax remains valid with minor updates	F08 — Python 3 Official Tutorial
P02	Learn Python - Full Course for Beginners (freeCodeCamp)	https://www.youtube.com/watch?v=eWRfhZUzrAc	D	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Single video chapter times can change with re-uploads; use YouTube's displayed chapters	F08 — Python 3 Official Tutorial
P03	Automate the Boring Stuff with Python	https://automatetheboringstuff.com/	D	Verified HTTP 200; endpoint reachable	Free online; paid print/course optional	2026-10-27	Some third-party APIs or screenshots are dated; use current library docs	F08 — Python 3 Official Tutorial
P04	RegexOne	https://regexone.com/	D	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Regex dialects differ; verify in Python re docs	F08 — Python 3 Official Tutorial
P05	pytest documentation	https://docs.pytest.org/en/stable/	D	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Plugin compatibility varies	F08 — Python 3 Official Tutorial
P06	Requests: HTTP for Humans	https://requests.readthedocs.io/en/latest/	D,F	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Never disable certificate verification except in a controlled local TLS lab	F10 — MDN HTTP Overview
P07	SQLite documentation	https://www.sqlite.org/docs.html	A,D	Verified HTTP 200; endpoint reachable	Free	2026-10-27	SQLite is not a substitute for learning PostgreSQL/MySQL administration	F11 — SQLBolt
P08	OWASP Secure Coding Practices Quick Reference Guide	https://owasp.org/www-project-secure-coding-practices-quick-reference-guide/	D,F	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Checklist, not a tutorial or complete standard	W01 — PortSwigger Web Security Academy
C01	Professor Messer SY0-701 Security+ Course Index	https://www.professormesser.com/security-plus/sy0-701/sy0-701-video/sy0-701-comptia-security-plus-course/	E,J	Verified HTTP 200; endpoint reachable	Free videos; paid notes/practice exams optional	2026-10-27	SY0-701 is V7; V8/SY0-801 launches Nov 17 2026, so use current objectives if certifying later	C02 — Professor Messer SY0-701 Security+ official playlist
C02	Professor Messer SY0-701 Security+ official playlist	https://www.youtube.com/playlist?list=PLG49S3nxzAnl4QDVqK-hOnoqcSKEIDDuv	E,J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	V8 content will differ after Nov 2026	C01 — Professor Messer SY0-701 Security+ Course Index
C03	ISC2 Certified in Cybersecurity Exam Outline	https://www.isc2.org/certifications/cc/cc-certification-exam-outline	E,J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Exam domains change; download current outline before paying	C04 — ISC2 CC self-study resources
C04	ISC2 CC self-study resources	https://www.isc2.org/certifications/cc/cc-self-study-resources	E,J	Verified HTTP 200; endpoint reachable	Free resources; official training paid	2026-10-27	Marketing pages may change; exam outline controls	C03 — ISC2 Certified in Cybersecurity Exam Outline

ID	Resource	URL	Topic	Verification status	Access	Date checked	Known limitation	Backup
C05	ISC2 exam pricing	https://www.isc2.org/register-for-exam/isc2-exam-pricing	J	Verified HTTP 200; endpoint reachable	Free to view; exam paid	2026-10-27	Local taxes and currencies vary; price may change	C03 — ISC2 Certified in Cybersecurity Exam Outline
C06	NIST Cybersecurity Framework 2.0	https://www.nist.gov/cyberframework	E,J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Framework is not law and not a certification	C21 — CIS Controls Version 8
C07	NIST SP 800-61 Revision 3	https://csrc.nist.gov/pubs/sp/800/61/r3/final	E,H,J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Guidance is advisory for US federal systems; other jurisdictions may differ	C06 — NIST Cybersecurity Framework 2.0
C08	NIST SP 800-63B	https://pages.nist.gov/800-63-4/sp800-63b.html	E,F	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Draft/revision status and version changes; use the page's current edition marker	C09 — Crypto 101
C09	Crypto 101	https://www.crypto101.io/	E	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Course is educational, not a substitute for vetted libraries or proofs	C10 — Stanford Cryptography I
C10	Stanford Cryptography I	https://www.coursera.org/learn/crypto	E	Verified HTTP 200; endpoint reachable	Free audit/lectures may be available; verified certificate and graded homework paid	2026-10-27	Older course; primitives remain important but verify current protocol guidance	C09 — Crypto 101
C11	A Graduate Course in Applied Cryptography	https://toc.cryptobook.us/	E	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Graduate-level; not a Month 5 mandatory read	C09 — Crypto 101
C12	Let's Encrypt documentation	https://letsencrypt.org/docs/	E,F	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Vendor-specific ACME details; general PKI principles still apply	C13 — RFC 8446: TLS 1.3
C13	RFC 8446: TLS 1.3	https://www.rfc-editor.org/rfc/rfc8446	E,F	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Dense standard; use simplified material first	C12 — Let's Encrypt documentation
C14	OWASP Top 10	https://owasp.org/www-project-top-ten/	F	Verified HTTP 200; endpoint reachable	Free	2026-10-27	2021 edition; check current release before citing year in interviews	W01 — PortSwigger Web Security Academy
C15	OWASP Application Security Verification Standard	https://owasp.org/www-project-application-security-verification-standard/	F	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Formal standard; not all levels apply to student projects	P08 — OWASP Secure Coding Practices Quick Reference Guide
C16	OWASP API Security Top 10	https://owasp.org/API-Security/editions/2023/en/0x11-t10/	F	Verified HTTP 200; endpoint reachable	Free	2026-10-27	2023 edition; verify current release later	C14 — OWASP Top 10
C17	OWASP Threat Modeling community page	https://owasp.org/www-community/Threat_Modeling	E,F,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Community page is an overview, not a complete methodology	C18 — MITRE ATT&CK
C18	MITRE ATT&CK	https://attack.mitre.org/	E,H,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Not a maturity model or proof of detection coverage	C19 — MITRE D3FEND
C19	MITRE D3FEND	https://d3fend.mitre.org/	E,H,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Research project; not all entries have turnkey implementations	C18 — MITRE ATT&CK

ID	Resource	URL	Topic	Verification status	Access	Date checked	Known limitation	Backup
C20	NIST NICE Workforce Framework	https://www.nist.gov/itl/applied-cybersecurity/nice	J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	US-origin framework; employers use many different vocabularies	C06 — NIST Cybersecurity Framework 2.0
C21	CIS Controls Version 8	https://www.cisecurity.org/controls/v8	E,J	Verified HTTP 200; endpoint reachable	Free overview; some tools paid	2026-10-27	Not regulation; legal obligations depend on jurisdiction	C06 — NIST Cybersecurity Framework 2.0
C22	NIST SP 800-53 Revision 5	https://csrc.nist.gov/pubs/sp/800/53/r5/upd1/final	J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Large catalog; use search and mappings	C06 — NIST Cybersecurity Framework 2.0
C23	ISO/IEC 27001 overview	https://www.iso.org/standard/27001	J	Automated checker blocked (HTTP 403); official/canonical URL; use backup if browser access fails	Page free; standard document paid	2026-10-27	Full standard is not freely distributable; do not rely on unauthorized copies	C06 — NIST Cybersecurity Framework 2.0
C24	GDPR full text (EUR-Lex)	https://eur-lex.europa.eu/eli/reg/2016/679/oj	J	Accepted/asynchronous response (HTTP 202); official legal text; use backup if browser fails	Free	2026-10-27	EU regulation, not global law; seek legal advice for actual obligations	C25 — NIST Privacy Framework
C25	NIST Privacy Framework	https://www.nist.gov/privacy-framework	J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Voluntary framework, not legal compliance proof	C06 — NIST Cybersecurity Framework 2.0
C26	NIST SP 800-30 Revision 1	https://csrc.nist.gov/pubs/sp/800/30/r1/final	J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	US federal guidance; terminology can be adapted	C06 — NIST Cybersecurity Framework 2.0
C27	NIST SP 800-34 Revision 1	https://csrc.nist.gov/pubs/sp/800/34/r1/final	J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	US federal guidance; adapt to local context	C06 — NIST Cybersecurity Framework 2.0
W01	PortSwigger Web Security Academy	https://portswigger.net/web-security	F	Verified HTTP 200; endpoint reachable	Free account required	2026-10-27	Labs require a free account and internet; do not target real public sites	W02 — PortSwigger Web Security Academy Learning Paths
W02	PortSwigger Web Security Academy Learning Paths	https://portswigger.net/web-security/learning-paths	F	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Path names and lab counts can change	W01 — PortSwigger Web Security Academy
W03	PortSwigger All Labs index	https://portswigger.net/web-security/all-labs	F	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Expert labs can be frustrating early; postpone them	W02 — PortSwigger Web Security Academy Learning Paths
W04	MDN HTTP documentation	https://developer.mozilla.org/en-US/docs/Web/HTTP	F	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Large documentation set; use site search and assigned pages	F10 — MDN HTTP Overview
W05	Burp Suite documentation	https://portswigger.net/burp/documentation	F,G	Verified HTTP 200; endpoint reachable	Community edition free; Pro paid	2026-10-27	Some features require paid Pro edition	W01 — PortSwigger Web Security Academy
W06	OWASP Web Security Testing Guide	https://owasp.org/www-project-web-security-testing-guide/	F,G	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Guide is not a step-by-step beginner tutorial	W01 — PortSwigger Web Security Academy
W07	OWASP ZAP documentation	https://www.zaproxy.org/docs/	F,G	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	Automated scans do not prove exploitability or absence of bugs	W05 — Burp Suite documentation

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W08	OWASP Juice Shop	https://owasp.org/www-project-juice-shop/	F,G	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	Never expose intentionally vulnerable software to the public internet	W09 — OWASP Juice Shop GitHub repository
W09	OWASP Juice Shop GitHub repository	https://github.com/juice-shop/juice-shop	F,G	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	Development branch may change; use official tags or documented release	W08 — OWASP Juice Shop
W10	DVWA GitHub repository	https://github.com/digininja/DVWA	F,G	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	Outdated interface and some PHP dependencies; do not expose publicly	W09 — OWASP Juice Shop GitHub repository
W11	OWASP WebGoat	https://owasp.org/www-project-webgoat/	F,G	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	Requires Java/container resources; use selectively	W01 — PortSwigger Web Security Academy
W12	OWASP Dependency-Check	https://owasp.org/www-project-dependency-check/	F,I	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	NVD API rate limits and database bootstrap can be slow; use offline sample report if blocked	W15 — OpenSSF Scorecard
W13	GitHub code security documentation	https://docs.github.com/en/code-security	F,I	Verified HTTP 200; endpoint reachable	Free features vary by repository visibility and plan	2026-10-27	Feature availability differs between public/private/enterprise plans	W14 — Semgrep documentation
W14	Semgrep documentation	https://semgrep.dev/docs/	F,I	Verified HTTP 200; endpoint reachable	Open-source engine free; some platform features paid	2026-10-27	Cloud platform and engine features differ	W13 — GitHub code security documentation
W15	OpenSSF Scorecard	https://securityscorecards.dev/	F,I	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	Not all checks apply to small student projects	W13 — GitHub code security documentation
W16	SLSA specification	https://slsa.dev/	F,I	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Specification evolves; use current level documentation	W13 — GitHub code security documentation
W17	OWASP Cheat Sheet Series	https://cheatsheetseries.owasp.org/	F	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Cheat sheets can be dense; use them after hands-on practice	C15 — OWASP Application Security Verification Standard
E01	NIST SP 800-115	https://csrc.nist.gov/pubs/sp/800/115/final	G	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Published 2008; methodology remains valid but tool examples are outdated	W06 — OWASP Web Security Testing Guide
E02	TryHackMe Pre Security learning path	https://tryhackme.com/path/outline/presecurity	A,B,E,G	Rate-limited automated check (HTTP 429); official URL; retry in browser or use backup	Freemium; introductory rooms free, full access may require subscription	2026-10-27	Free-room availability and path structure changed in 2026; verify before buying	L01 — Linux Journey
E03	TryHackMe Cyber Security 101 path	https://tryhackme.com/path/outline/cybersecurity101	E,G	Rate-limited automated check (HTTP 429); official URL; retry in browser or use backup	Freemium; many rooms subscription-only	2026-10-27	Path content and free/paid status can change	E01 — NIST SP 800-115
E04	Hack The Box Academy	https://academy.hackthebox.com/	G,K	Verified HTTP 200; endpoint reachable	Freemium; some modules free, others require cubes/subscription	2026-10-27	Pricing and free-module availability vary	N09 — Nmap Reference Guide

ID	Resource	URL	Topic	Verification status	Access	Date checked	Known limitation	Backup
E05	HackerOne Disclosure Guidelines	https://www.hackerone.com/disclosure-guidelines	G,J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Platform policy is not universal law and does not grant authorization for arbitrary targets	E06 — FIRST CVSS specification
E06	FIRST CVSS specification	https://www.first.org/cvss/	G,J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	CVSS is not a full risk assessment	E08 — NVD
E07	CVE Program	https://www.cve.org/	G,J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	CVE assignment itself does not prove exploitability or priority	E08 — NVD
E08	NVD	https://nvd.nist.gov/	G,J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	NVD enrichment can lag; API rate limits apply	N09 — Nmap Reference Guide
E09	picoCTF	https://picoctf.org/	D,G,K	Verified HTTP 200; endpoint reachable	Free competition/ picoGym account	2026-10-27	Competition archive availability and scoreboards change	L15 — pwn.college Dojo index
E10	CyLab Security Academy / picoGym	https://www.cylabacademy.org/	D,G,K	Verified HTTP 200; endpoint reachable	Free account	2026-10-27	Practice catalog may change	E09 — picoCTF
B01	Microsoft Sentinel documentation	https://learn.microsoft.com/en-us/azure/sentinel/	H	Verified HTTP 200; endpoint reachable	Docs free; Azure/ Sentinel usage may incur costs	2026-10-27	Product names and UI change; cloud cost risk	B02 — Kusto Query Language overview
B02	Kusto Query Language overview	https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric	H	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Syntax and service-specific tables vary by product	B03 — SigmaHQ
B03	SigmaHQ	https://sigmahq.io/	H,K	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	Rule quality varies; always validate in your own data	B18 — CISA Known Exploited Vulnerabilities Catalog
B04	Sigma rule creation guide	https://sigmahq.io/docs/guide/getting-started.html	H,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Documentation URL or schema can change; use site navigation	B03 — SigmaHQ
B05	SBousseaden EVTX-ATTACK-SAMPLES	https://github.com/sbousseaden/EVTX-ATTACK-SAMPLES	H,K	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	Third-party repository; verify contents and do not execute unknown binaries; EVTX files are data	B02 — Kusto Query Language overview
B06	CyberDefenders	https://cyberdefenders.org/	H,K	Verified HTTP 200; endpoint reachable	Freemium; some challenges free, some require subscription	2026-10-27	Free/paid status changes; some datasets are large	N07 — Wireshark SampleCaptures
B07	Blue Team Labs Online	https://blueteamlabs.online/	H,K	Verified HTTP 200; endpoint reachable	Freemium; free and paid labs	2026-10-27	Paid tier/lab catalogue changes; lab platform login required	B06 — CyberDefenders
B08	Malware Traffic Analysis	https://malware-traffic-analysis.net/	B,H,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Contains malware-related artifacts; never execute samples; some links may be old	N07 — Wireshark SampleCaptures
B09	Zeek documentation	https://docs.zeek.org/en/master/	B,H	Rate-limited automated check (HTTP 429);	Free/open-source	2026-10-27	Installation and version differences; not needed on low-end hardware	B10 — Suricata documentation

ID	Resource	URL	Topic	Verification status	Access	Date checked	Known limitation	Backup
				official URL; retry in browser or use backup				
B10	Suricata documentation	https://docs.suricata.io/en/latest/	B,H	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	Rule syntax and version-specific features; use matching docs version	B03 — SigmaHQ
B11	Autopsy documentation	https://www.autopsy.com/support/	H,K	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	Older screenshots may differ from current release; never analyze personal disks without permission	B12 — SANS SIFT Workstation
B12	SANS SIFT Workstation	https://www.sans.org/tools/sift-workstation/	H,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Large download and Linux familiarity required; SANS courses themselves are paid	B11 — Autopsy documentation
B13	MISP Project	https://www.misp-project.org/	H,K	Verified HTTP 200; endpoint reachable	Free/open-source; hosting costs possible	2026-10-27	Do not upload personal or unauthorized data; local hosting can be heavy	C18 — MITRE ATT&CK
B14	Elastic detection rules documentation	https://www.elastic.co/guide/en/security/current/detection-engine-overview.html	H,K	Verified HTTP 200; endpoint reachable	Free docs/open-source rules; Elastic Stack may have paid features	2026-10-27	Product UI and rule schema change by version	B03 — SigmaHQ
B15	Microsoft SC-200 certification page	https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/	H,J	Verified HTTP 200; endpoint reachable	Free learning content; exam price region-based	2026-10-27	Exam objectives were refreshed in 2026; check current study guide	B01 — Microsoft Sentinel documentation
B16	Blue Team Level 1 official certification	https://www.centri.org/certifications/blue-team-level-1	H,J	Verified HTTP 200; endpoint reachable	Paid; GBP 399 listed on official page as checked 2026-10-27; one free resit included	2026-10-27	Price, provider brand and access terms may change; not required for this roadmap	B06 — CyberDefenders
B17	CISA cybersecurity advisories	https://www.cisa.gov/news-events/cybersecurity-advisories	H,J,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	US government source; not exhaustive global coverage	C18 — MITRE ATT&CK
B18	CISA Known Exploited Vulnerabilities Catalog	https://www.cisa.gov/known-exploited-vulnerabilities-catalog	G,J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	US federal remediation deadlines do not automatically bind Pakistani or private organizations	E08 — NVD
B19	SC-200 Exam Prep YouTube playlist	https://www.youtube.com/playlist?list=PLahhVEj9XNTcHNtYH-inn7xFJactEM7zz	H,J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Certification content can change; use current study guide first	B15 — Microsoft SC-200 certification page
CL01	AWS Skill Builder	https://skillbuilder.aws/	I,J	Verified HTTP 200; endpoint reachable	Freemium; many digital courses free, labs/ subscription can be paid	2026-10-27	Catalog and free/paid status changes	CL02 — AWS security documentation

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CL02	AWS security documentation	https://docs.aws.amazon.com/security/	I	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Very broad; use search rather than linear reading	CL03 — AWS Well-Architected Security Pillar
CL03	AWS Well-Architected Security Pillar	https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html	I,J	Verified HTTP 200; endpoint reachable	Free	2026-10-27	AWS-specific; other clouds need their own architecture guidance	CL02 — AWS security documentation
CL04	AWS Free Tier	https://aws.amazon.com/free/	I	Verified HTTP 200; endpoint reachable	Program terms; some services free temporarily, others always free	2026-10-27	Terms change; credits/expiry vary; a credit card may still be required	CL01 — AWS Skill Builder
CL05	Microsoft Azure security documentation	https://learn.microsoft.com/en-us/azure/security/	I	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Product names and services change	CL02 — AWS security documentation
CL06	Docker documentation	https://docs.docker.com/	C,I	Verified HTTP 200; endpoint reachable	Free; Docker Desktop licensing may apply in some organizations	2026-10-27	Menu changes and Docker Desktop license terms differ from Docker Engine	CL07 — Docker Engine security documentation
CL07	Docker Engine security documentation	https://docs.docker.com/engine/security/	I	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Host kernel distribution affects defaults	CL06 — Docker documentation
CL08	Kubernetes official documentation	https://kubernetes.io/docs/home/	I	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Version-specific APIs; use supported version docs	CL09 — Kubernetes security documentation
CL09	Kubernetes security documentation	https://kubernetes.io/docs/concepts/security/	I	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Some features require cluster configuration that kind/minikube may not fully emulate	CL08 — Kubernetes official documentation
CL10	OWASP Kubernetes Top 10	https://owasp.org/www-project-kubernetes-top-ten/	I	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Checklist, not a deployment tutorial	CL09 — Kubernetes security documentation
CL11	Kubernetes Goat	https://madhuakula.com/kubernetes-goat/	I,K	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	Can consume significant RAM; skip if hardware is below 16 GB	CL10 — OWASP Kubernetes Top 10
CL12	Trivy documentation	https://trivy.dev/docs/	I	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	Vulnerability DB freshness and false positives; no scanner proves security	W12 — OWASP Dependency-Check
CL13	Checkov documentation	https://www.checkov.io/1.Welcome/What%20is%20Checkov.html	I	Verified HTTP 200; endpoint reachable	Free/open-source; Prisma Cloud features paid	2026-10-27	Policy coverage varies by framework and version	CL12 — Trivy documentation
CL14	GitHub Actions security hardening	https://docs.github.com/en/actions/security-for-github-actions/security-guides/security-	I	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Features differ for public/private/enterprise repos	W13 — GitHub code security documentation

ID	Resource	URL	Topic	Verification status	Access	Date checked	Known limitation	Backup
		hardening-for-github-actions						
CL15	Terraform documentation	https://developer.hashicorp.com/terraform/docs	I	Verified HTTP 200; endpoint reachable	Free/open-source; HCP Terraform has paid tiers	2026-10-27	State files can contain secrets; never commit them	CL13 — Checkov documentation
CL16	Cloud Security Alliance Cloud Controls Matrix	https://cloudsecurityalliance.org/research/cloud-controls-matrix	I,J	Verified HTTP 200; endpoint reachable	Free registration may be required for some downloads	2026-10-27	Registration and version changes; not law	CL03 — AWS Well-Architected Security Pillar
CL17	AWS Certified Security - Specialty official page	https://aws.amazon.com/certification/certified-security-specialty/	I,J	Verified HTTP 200; endpoint reachable	Exam US$300 listed 2026-10-27; training optional/paid	2026-10-27	Region/language and recertification policies change; not entry-level	CL01 — AWS Skill Builder
CL18	AWS recertification page	https://aws.amazon.com/certification/recertification/	J	Verified HTTP 200; endpoint reachable	Free to read; exam/maintenance may cost	2026-10-27	Two official AWS pages describe policy in different detail; verify in your certification account	CL17 — AWS Certified Security - Specialty official page
A01	pwn.college	https://pwn.college/	A,C,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Course instance names change yearly; use Core Material	L15 — pwn.college Dojo index
A02	pwn.college dojos index	https://pwn.college/dojos	A,C,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Dojo catalogue changes; links may require login	A01 — pwn.college
A03	OpenSecurityTraining2	https://ost2.fyi/	A,K	Verified HTTP 200; endpoint reachable	Free/open-source course materials	2026-10-27	Many classes are long and assume C; do not attempt in Month 1	A01 — pwn.college
A04	OST2 Architecture 1001: x86-64 Assembly	https://p.ost2.fyi/courses/course-v1:OpenSecurityTraining2+Arch1001_x86-64_Asm+2021_v1/about	A,K	Verified HTTP 200; endpoint reachable	Free; optional certificate mechanics may vary	2026-10-27	Older recordings but durable architecture concepts; verify tools on current OS	A03 — OpenSecurityTraining2
A05	Malware Unicorn Reverse Engineering 101	https://malwareunicorn.org/workshops/re101.html	K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	2019 VM images and tools are old; use current official installers and isolated networking	A04 — OST2 Architecture 1001: x86-64 Assembly
A06	MIT 6.858 Computer Systems Security OCW	https://ocw.mit.edu/courses/6-858-computer-systems-security-fall-2014/	E,F,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	2014 material; timeless foundations but obsolete tool/browser specifics	A07 — MIT 6.858 official lecture playlist
A07	MIT 6.858 official lecture playlist	https://www.youtube.com/playlist?list=PLUl4u3cNGP62K2DjQLRxDNRi0z2IRWnNh	E,F,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	2014 publication; update all practical tool details	A06 — MIT 6.858 Computer Systems Security OCW
A08	Security Engineering, Third Edition (free online chapters)	https://www.cl.cam.ac.uk/~rja14/book.html	E,J,K	Verified HTTP 200; endpoint reachable	Free online draft; print edition paid	2026-10-27	Third-edition draft chapters may be updated; not a certification cram book	C06 — NIST Cybersecurity Framework 2.0

ID	Resource	URL	Topic	Verification status	Access	Date checked	Known limitation	Backup
A09	Nightmare binary exploitation course	https://guyinatuxedo.github.io/	K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Advanced and sometimes terse; not a Year 1 mastery target	A01 — pwn.college
A10	RPISEC Modern Binary Exploitation	https://github.com/RPISEC/MBE	K	Verified HTTP 200; endpoint reachable	Free/open-source	2026-10-27	Older VM images and mitigations; modern mitigations require updated setup	A01 — pwn.college
A11	Google Project Zero blog	https://googleprojectzero.blogspot.com/	K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Advanced; many posts require specialized background	A12 — USENIX Security conference papers
A12	USENIX Security conference papers	https://www.usenix.org/conferences/byname/164	K	Verified HTTP 200; endpoint reachable	Many papers freely available; conference attendance paid	2026-10-27	Not every paper includes code; reproduction may be hard	A11 — Google Project Zero blog
A13	arXiv cs.CR recent papers	https://arxiv.org/list/cs.CR/recent	K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Preprints are not necessarily peer reviewed; quality varies	A12 — USENIX Security conference papers
A14	xairy/linux-kernel-exploitation	https://github.com/xairy/linux-kernel-exploitation	K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Dangerous material if misunderstood; use isolated research VMs only	A01 — pwn.college
A15	OWASP Top 10 for LLM Applications	https://genai.owasp.org/llm-top-10/	F,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Versioned document; verify current edition before citing	C16 — OWASP API Security Top 10
A16	NIST AI Risk Management Framework	https://www.nist.gov/itl/ai-risk-management-framework	J,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Not law and not an appsec testing checklist	A15 — OWASP Top 10 for LLM Applications
A17	Cisco CCST Cybersecurity official exam page	https://www.cisco.com/site/us/en/learn/training-certifications/exams/ccst-cybersecurity.html	B,E,J	Verified HTTP 200; endpoint reachable	Exam paid US$125 listed; free Cisco training linked	2026-10-27	Cisco certification names/pathways are changing in 2026; verify current page and 5-year renewal rules	E01 — NIST SP 800-115
A18	CompTIA Security+ V7 and V8 official pages	https://www.comptia.org/en-us/certifications/security/	E,J	Verified HTTP 200; endpoint reachable	Paid exam; V7 US$404 from CompTIA blog dated January 2025; verify current store	2026-10-27	V7 retirement English 2027-06-11; V8 launch November 2026; do not buy stale practice material	C01 — Professor Messer SY0-701 Security+ Course Index
A19	ISC2 Certified in Cybersecurity official page	https://www.isc2.org/Certifications/CC	E,J	Verified HTTP 200; endpoint reachable	Exam paid; US$199 listed on official pricing page 2026-10-27; occasional free promotions may exist	2026-10-27	Annual maintenance fee after certification; exact local price/tax at checkout	C03 — ISC2 Certified in Cybersecurity Exam Outline
A20	Microsoft SC-900 official certification page	https://learn.microsoft.com/en-us/credentials/certifications/security-compliance-and-identity-fundamentals/	E,J	Verified HTTP 200; endpoint reachable	Free Learn training; exam price region-based	2026-10-27	Price at checkout depends on country; event vouchers are not guaranteed	B15 — Microsoft SC-200 certification page
A21			G,K			2026-10-27		E01 — NIST SP 800-115

ID	Resource	URL	Topic	Verification status	Access	Date checked	Known limitation	Backup
	OffSec PEN-200 / OSCP+ official page	https://www.offsec.com/courses/pen-200/		Verified HTTP 200; endpoint reachable	Paid; official page lists US$1749 course+exam bundle and US$1699 standalone exam on 2026-10-27		High cost, renewal/maintenance conditions and exam format can change; free alternatives exist	
A22	INE eJPT official certification page	https://ine.com/security/certifications/ejpt-certification	G,K	Verified HTTP 200; endpoint reachable	Paid; official page states an INE subscription and exam voucher are needed; checkout lists voucher/bundle prices	2026-10-27	Official pricing is split between subscription and voucher; checkout promotions change; certification valid three years	E01 — NIST SP 800-115
A23	KQL Essentials official Azure Data Explorer playlist	https://www.youtube.com/playlist?list=PLKHIhZJiQ5AkOMEC54mQQcAkfg4UXT8uK	H	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Individual publication dates vary; services and UI can change	B02 — Kusto Query Language overview
A24	SANS DFIR playlist	https://www.youtube.com/playlist?list=PLrY_AbzZGqt9MJiLjyH-zTrypGLs154e4	H,K	Verified HTTP 200; endpoint reachable	Free	2026-10-27	Some SANS courses are paid; webcasts may be older; do not confuse a webcast with certification	B11 — Autopsy documentation

Detailed resource cards (R001-R169)

F01 · CS50x 2026 OpenCourseWare

Exact title	CS50x 2026 OpenCourseWare
Direct URL	https://cs50.harvard.edu/x/
Resource type / domains	University course · Modules A,D
Topic and subtopic coverage	Best single academic bridge from programming and CS fundamentals to Python, SQL, web and systems thinking
Exact path / module / chapter / playlist	Weeks 0-10; problem sets; final project
Recommended difficulty / estimated time	Beginner to intermediate · 11 weeks part-time
Access status	Free; paid edX verified certificate optional
Prerequisites	Basic computer literacy
Why selected	Best single academic bridge from programming and CS fundamentals to Python, SQL, web and systems thinking
What to practise after	Submit selected problem sets from Weeks 0,1,4,6,7,9 as scheduled in the daily plan
Backup if inaccessible	F02 — CS50x Lectures 2026 (official YouTube playlist)
Known limitations	Annual edition changes; use 2026 or current site
Classification / age status	Mandatory · Current 2026
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

F02 · CS50x Lectures 2026 (official YouTube playlist)

Exact title	CS50x Lectures 2026 (official YouTube playlist)
Direct URL	https://www.youtube.com/playlist?list=PLhQjrBD2T380hlTqAU8HfvVepCcjCqTg6
Resource type / domains	YouTube playlist · Modules A,D
Topic and subtopic coverage	Harvard's current lecture sequence with chapters and transcripts on the course site
Exact path / module / chapter / playlist	Lecture 0 through Lecture 10; use exact lecture titles shown on YouTube
Recommended difficulty / estimated time	Beginner to intermediate · 25 hours video
Access status	Free
Prerequisites	None
Why selected	Harvard's current lecture sequence with chapters and transcripts on the course site
What to practise after	Watch one lecture segment, then reproduce every demonstrated command or code snippet
Backup if inaccessible	F01 — CS50x 2026 OpenCourseWare
Known limitations	Playlist titles can be renumbered annually; match the official week page first
Classification / age status	Recommended · 2026
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

F03 · CS50x Sections 2026 (official YouTube playlist)

Exact title	CS50x Sections 2026 (official YouTube playlist)
Direct URL	https://www.youtube.com/playlist?list=PLhQjrBD2T3839iqxqUdw7NWOCboh00vsV
Resource type / domains	YouTube playlist · Modules A,D
Topic and subtopic coverage	Short worked examples make abstract lectures easier to translate into code
Exact path / module / chapter / playlist	Sections: C, Arrays, Algorithms, Memory, Data Structures, Python, SQL, HTML/CSS/JS, Flask
Recommended difficulty / estimated time	Beginner to intermediate · 12-18 hours
Access status	Free
Prerequisites	CS50 lecture or equivalent
Why selected	Short worked examples make abstract lectures easier to translate into code
What to practise after	Code along without looking, then delete and recreate from memory
Backup if inaccessible	F02 — CS50x Lectures 2026 (official YouTube playlist)
Known limitations	Older section videos may use previous tool versions
Classification / age status	Optional enrichment · 2026
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

F04 · CS50's Introduction to Cybersecurity

Exact title	CS50's Introduction to Cybersecurity
Direct URL	https://cs50.harvard.edu/cybersecurity/
Resource type / domains	University course · Modules E
Topic and subtopic coverage	Risk-based, technically honest introduction to security trade-offs
Exact path / module / chapter / playlist	Weeks 0-4: Securing Accounts, Data, Systems, Software, Privacy; final project
Recommended difficulty / estimated time	Beginner · 5 weeks at 2-6 h/week
Access status	Free OCW; paid edX verified certificate optional
Prerequisites	None
Why selected	Risk-based, technically honest introduction to security trade-offs
What to practise after	Complete each week's assignment and one final project; do not use it as a substitute for hands-on labs
Backup if inaccessible	F05 — CS50's Introduction to Cybersecurity (official YouTube playlist)
Known limitations	High level; does not teach SOC tooling, exploitation, or secure coding in depth
Classification / age status	Mandatory · Current site; 2023 lecture recordings
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

F05 · CS50's Introduction to Cybersecurity (official YouTube playlist)

Exact title	CS50's Introduction to Cybersecurity (official YouTube playlist)
Direct URL	https://www.youtube.com/playlist?list=PLhQjrBD2T383Cqo5I1oRrbC1EKRAKGKUE
Resource type / domains	YouTube playlist · Modules E
Topic and subtopic coverage	Easy to watch in segments and cross-reference with official notes
Exact path / module / chapter / playlist	Five lectures corresponding to Weeks 0-4
Recommended difficulty / estimated time	Beginner · 8-10 hours
Access status	Free
Prerequisites	None
Why selected	Easy to watch in segments and cross-reference with official notes
What to practise after	After each lecture, write a one-page risk trade-off analysis
Backup if inaccessible	F04 — CS50's Introduction to Cybersecurity
Known limitations	2023 publication date; core concepts remain valid, current product interfaces may differ
Classification / age status	Recommended · Old but gold (2023)
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

F06 · Operating Systems: Three Easy Pieces (OSTEP)

Exact title	Operating Systems: Three Easy Pieces (OSTEP)
Direct URL	https://pages.cs.wisc.edu/~remzi/OSTEP/
Resource type / domains	Free textbook · Modules A,C
Topic and subtopic coverage	University-grade explanation of processes, virtualization, concurrency, persistence and security fundamentals
Exact path / module / chapter / playlist	Chapters on Processes, Process API, Direct Execution, Scheduling, Address Spaces, Memory API, Translation, Files and Directories, Security chapters
Recommended difficulty / estimated time	Intermediate · Selected chapters 15-25 h
Access status	Free online chapters; paid print optional
Prerequisites	Basic C/Python and computer architecture
Why selected	University-grade explanation of processes, virtualization, concurrency, persistence and security fundamentals
What to practise after	Draw mechanism diagrams and answer end-of-chapter questions; reproduce tiny C examples where available
Backup if inaccessible	F01 — CS50x 2026 OpenCourseWare
Known limitations	Some examples assume Unix/C familiarity; do not attempt every chapter in Month 1
Classification / age status	Recommended · v1.10, 2023
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

F07 · Nand2Tetris / The Elements of Computing Systems

Exact title	Nand2Tetris / The Elements of Computing Systems
Direct URL	https://www.nand2tetris.org/
Resource type / domains	Free course and software · Modules A
Topic and subtopic coverage	Builds a concrete mental model of how logic gates become a CPU
Exact path / module / chapter / playlist	Part I projects 1-5: Boolean logic, Boolean arithmetic, sequential logic, machine language, computer architecture
Recommended difficulty / estimated time	Intermediate · 20-40 h for Part I
Access status	Free software and course materials; book optional paid
Prerequisites	Basic programming and binary arithmetic
Why selected	Builds a concrete mental model of how logic gates become a CPU
What to practise after	Complete Projects 1-3 or a software simulation if hardware simulators are blocked
Backup if inaccessible	F06 — Operating Systems: Three Easy Pieces (OSTEP)
Known limitations	Large time commitment; use selected projects only
Classification / age status	Optional enrichment · Classic, technically durable
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

F08 · Python 3 Official Tutorial

Exact title	Python 3 Official Tutorial
Direct URL	https://docs.python.org/3/tutorial/
Resource type / domains	Official documentation · Modules A,D
Topic and subtopic coverage	Canonical, current Python syntax and standard library semantics
Exact path / module / chapter / playlist	Sections 1-10: interpreter, control flow, functions, data structures, modules, input/output, errors, classes, standard library
Recommended difficulty / estimated time	Beginner to intermediate · 12-18 h
Access status	Free
Prerequisites	None
Why selected	Canonical, current Python syntax and standard library semantics
What to practise after	Type each example locally, then write a variation without copying
Backup if inaccessible	P02 — Learn Python - Full Course for Beginners (freeCodeCamp)
Known limitations	Reference style can be terse for absolute beginners
Classification / age status	Mandatory · Current documentation
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

F09 · Pro Git, 2nd Edition

Exact title	Pro Git, 2nd Edition
Direct URL	https://git-scm.com/book/en/v2
Resource type / domains	Free book / official reference · Modules A,D
Topic and subtopic coverage	Authoritative Git model rather than memorized button clicks
Exact path / module / chapter / playlist	Chapters 1-3 and 7: getting started, basics, branching, selected tools
Recommended difficulty / estimated time	Beginner to intermediate · 8-12 h
Access status	Free
Prerequisites	Command line basics
Why selected	Authoritative Git model rather than memorized button clicks
What to practise after	Create a repository, branch, merge, resolve a deliberate conflict, and publish a clean README
Backup if inaccessible	P01 — Python for Everybody
Known limitations	GUI screenshots may differ from local tools
Classification / age status	Mandatory · Classic, current model
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

F10 · MDN HTTP Overview

Exact title	MDN HTTP Overview
Direct URL	https://developer.mozilla.org/en-US/docs/Web/HTTP/Guides/Overview
Resource type / domains	Official web documentation · Modules A,B,F
Topic and subtopic coverage	Clear browser-vendor documentation for HTTP semantics and security-relevant headers
Exact path / module / chapter / playlist	HTTP basics, requests, responses, methods, headers, status codes, cookies, caching
Recommended difficulty / estimated time	Beginner · 2-4 h
Access status	Free
Prerequisites	Basic networking vocabulary
Why selected	Clear browser-vendor documentation for HTTP semantics and security-relevant headers
What to practise after	Inspect requests in browser DevTools and map each header to MDN
Backup if inaccessible	W04 — MDN HTTP documentation
Known limitations	Some pages move as MDN reorganizes; use site search if a URL changes
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

F11 · SQLBolt

Exact title	SQLBolt
Direct URL	https://sqlbolt.com/
Resource type / domains	Interactive lessons · Modules A,D
Topic and subtopic coverage	Short, interactive SQL drills with immediate feedback suitable for a first database foundation
Exact path / module / chapter / playlist	Lessons 1-18: SELECT, filtering, joins, aggregation, order, inserts, updates, deletes, schema
Recommended difficulty / estimated time	Beginner · 4-8 h
Access status	Free
Prerequisites	None
Why selected	Short, interactive SQL drills with immediate feedback suitable for a first database foundation
What to practise after	Build a synthetic security-events table and write SELECT, WHERE, GROUP BY, JOIN and UPDATE exercises
Backup if inaccessible	P08 — OWASP Secure Coding Practices Quick Reference Guide
Known limitations	SQLite dialect differs slightly from PostgreSQL or SQL Server
Classification / age status	Recommended · Classic, technically durable
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

F12 · Oracle VirtualBox User Manual

Exact title	Oracle VirtualBox User Manual
Direct URL	https://www.virtualbox.org/manual/
Resource type / domains	Official documentation · Modules A,C
Topic and subtopic coverage	Authoritative setup and isolation reference for a no-cost lab
Exact path / module / chapter / playlist	Chapters 1-3, 5-6, 8: introduction, installation, basic configuration, virtual storage, networking, snapshots
Recommended difficulty / estimated time	Beginner · Selected sections 4-6 h
Access status	Free
Prerequisites	Desktop OS administration basics
Why selected	Authoritative setup and isolation reference for a no-cost lab
What to practise after	Create an Ubuntu VM with NAT plus an isolated internal adapter and take a clean snapshot
Backup if inaccessible	L07 — Microsoft Windows Server documentation
Known limitations	Version-specific menus may differ; use the manual matching your installed version
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

F13 · Ubuntu Desktop Guide

Exact title	Ubuntu Desktop Guide
Direct URL	https://help.ubuntu.com/
Resource type / domains	Official documentation · Modules A,C
Topic and subtopic coverage	Stable first Linux desktop reference
Exact path / module / chapter / playlist	Installation, desktop, files, networking, users, software, troubleshooting
Recommended difficulty / estimated time	Beginner · Selected sections 3-6 h
Access status	Free
Prerequisites	None
Why selected	Stable first Linux desktop reference
What to practise after	Configure users, package updates, firewall and logs in the lab VM
Backup if inaccessible	L01 — Linux Journey
Known limitations	Release-specific UI changes
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

F14 · Microsoft Learn Windows fundamentals

Exact title	Microsoft Learn Windows fundamentals
Direct URL	https://learn.microsoft.com/en-us/training/modules/explore-windows-client/
Resource type / domains	Official learning hub · Modules A,C
Topic and subtopic coverage	Structured Microsoft-authored introduction before Active Directory
Exact path / module / chapter / playlist	Windows client and server administration learning paths; select modules on accounts, processes, services and PowerShell
Recommended difficulty / estimated time	Beginner · 8-20 h
Access status	Free learning content; exams separate
Prerequisites	A Windows lab VM or Windows host
Why selected	Structured Microsoft-authored introduction before Active Directory
What to practise after	Document processes, services, permissions and event IDs from your own VM
Backup if inaccessible	L08 — PowerShell documentation
Known limitations	Product names and UI change; rely on current Microsoft Learn pages
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N01 · Jeremy's IT Lab Free CCNA 200-301 Complete Course

Exact title	Jeremy's IT Lab Free CCNA 200-301 Complete Course
Direct URL	https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ
Resource type / domains	YouTube playlist · Modules B
Topic and subtopic coverage	Exceptional depth and free Packet Tracer labs for real networking fundamentals
Exact path / module / chapter / playlist	Days 1-63; begin with Network Devices, Interfaces and Cables, TCP/IP Model, CLI, Ethernet LAN Switching, IPv4 Addressing
Recommended difficulty / estimated time	Beginner to intermediate · 63 study days; selected first 45 days in this roadmap
Access status	Free on YouTube; optional paid course extras
Prerequisites	None
Why selected	Exceptional depth and free Packet Tracer labs for real networking fundamentals
What to practise after	Watch one day, complete the matching Packet Tracer lab and Anki cards where accessible
Backup if inaccessible	N02 — Jeremy's IT Lab CCNA Course Index and Files
Known limitations	Playlist originated in 2020 and was augmented for v1.1; verify current Cisco blueprint before certification study
Classification / age status	Mandatory · Old but gold, updated additions
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N02 · Jeremy's IT Lab CCNA Course Index and Files

Exact title	Jeremy's IT Lab CCNA Course Index and Files
Direct URL	https://courses.jeremysitlab.com/p/ccna
Resource type / domains	Course index and lab files · Modules B
Topic and subtopic coverage	The index supplies verifiable lesson titles and lab names without inventing timestamps
Exact path / module / chapter / playlist	Course Content: Days 1-63; lecture titles, durations, labs; free files link via video descriptions
Recommended difficulty / estimated time	Beginner to intermediate · Same as N01
Access status	Free YouTube course; paid ad-free edition has bonus quizzes
Prerequisites	None
Why selected	The index supplies verifiable lesson titles and lab names without inventing timestamps
What to practise after	Use the matching Day lecture title and Packet Tracer lab; download files only from the official link
Backup if inaccessible	N03 — Jeremy's IT Lab CCNA Course Files
Known limitations	Paid edition is optional; free files are linked separately
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N03 · Jeremy's IT Lab CCNA Course Files

Exact title	Jeremy's IT Lab CCNA Course Files
Direct URL	https://jitl.jp/ccna-files
Resource type / domains	Downloadable labs/flashcards · Modules B
Topic and subtopic coverage	Official free practice material accompanying N01
Exact path / module / chapter / playlist	Anki decks and Packet Tracer lab files by course day
Recommended difficulty / estimated time	Beginner to intermediate · Varies
Access status	Free
Prerequisites	Packet Tracer and Anki
Why selected	Official free practice material accompanying N01
What to practise after	Complete the current Day's lab, then reconfigure it from a blank topology
Backup if inaccessible	N02 — Jeremy's IT Lab CCNA Course Index and Files
Known limitations	External host may change; use N02 video description if unavailable
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N04 · Professor Messer N10-009 Network+ Training Course

Exact title	Professor Messer N10-009 Network+ Training Course
Direct URL	https://www.youtube.com/playlist?list=PLG49S3nxzAnl_tQe3kvnmeMid0mjF8Le8
Resource type / domains	YouTube playlist · Modules B
Topic and subtopic coverage	Objective-indexed short videos are ideal for targeted review and gap filling
Exact path / module / chapter / playlist	Official 87-video N10-009 course; use exact numbered objective titles
Recommended difficulty / estimated time	Beginner to intermediate · 12 h 55 m total video
Access status	Free
Prerequisites	Basic IT vocabulary
Why selected	Objective-indexed short videos are ideal for targeted review and gap filling
What to practise after	After a Jeremy's IT Lab lesson, watch one Messer objective video as an alternate explanation
Backup if inaccessible	N05 — Professor Messer N10-009 Course Index
Known limitations	Certification-oriented, not a complete packet-analysis course
Classification / age status	Recommended · 2024 onward
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N05 · Professor Messer N10-009 Course Index

Exact title	Professor Messer N10-009 Course Index
Direct URL	https://www.professormesser.com/network-plus/n10-009/n10-009-video/n10-009-training-course/
Resource type / domains	Course index · Modules B
Topic and subtopic coverage	More reliable navigation than a channel homepage; exact video titles and run times
Exact path / module / chapter / playlist	All N10-009 video titles grouped by exam objective
Recommended difficulty / estimated time	Beginner to intermediate · 12 h 55 m total video
Access status	Free
Prerequisites	None
Why selected	More reliable navigation than a channel homepage; exact video titles and run times
What to practise after	Find the exact objective from the daily plan and open the linked video
Backup if inaccessible	N04 — Professor Messer N10-009 Network+ Training Course
Known limitations	Older N10-008 course also exists; ensure N10-009
Classification / age status	Recommended · 2024 onward
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N06 · Wireshark User's Guide

Exact title	Wireshark User's Guide
Direct URL	https://www.wireshark.org/docs/wsug_html_chunked/
Resource type / domains	Official documentation · Modules B,H
Topic and subtopic coverage	Primary reference for legally captured packets and repeatable analysis
Exact path / module / chapter / playlist	Chapters 3-7: user interface, capturing, display filters, coloring, protocol dissection, statistics
Recommended difficulty / estimated time	Beginner to intermediate · 8-12 h selected
Access status	Free
Prerequisites	Basic TCP/IP
Why selected	Primary reference for legally captured packets and repeatable analysis
What to practise after	Use display filters, follow streams, export objects and annotate packet fields in assigned sample captures
Backup if inaccessible	N07 — Wireshark SampleCaptures
Known limitations	Version-specific UI; match your installed Wireshark version
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N07 · Wireshark SampleCaptures

Exact title	Wireshark SampleCaptures
Direct URL	https://wiki.wireshark.org/SampleCaptures
Resource type / domains	Packet capture repository · Modules B,H
Topic and subtopic coverage	Provides legal, public captures for packet analysis without sniffing other people's traffic
Exact path / module / chapter / playlist	Sample captures by protocol; DNS, HTTP, TCP, TLS, ARP, ICMP and malware traffic
Recommended difficulty / estimated time	Intermediate · 1-3 h per capture
Access status	Free
Prerequisites	Wireshark fundamentals
Why selected	Provides legal, public captures for packet analysis without sniffing other people's traffic
What to practise after	Write a structured report: endpoints, conversations, protocol layers, anomaly, limitations
Backup if inaccessible	N06 — Wireshark User's Guide
Known limitations	Some linked external captures disappear; download from stable Wireshark-hosted mirrors where possible
Classification / age status	Mandatory · Maintained wiki with older captures
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N08 · tcpdump manual page

Exact title	tcpdump manual page
Direct URL	https://www.tcpdump.org/manpages/tcpdump.1.html
Resource type / domains	Official command reference · Modules B,H
Topic and subtopic coverage	Canonical syntax for command-line capture filters and safe packet collection
Exact path / module / chapter / playlist	Options, expression syntax, output formats, examples
Recommended difficulty / estimated time	Intermediate · 2-4 h
Access status	Free
Prerequisites	Linux command line and TCP/IP
Why selected	Canonical syntax for command-line capture filters and safe packet collection
What to practise after	Capture only loopback or own lab traffic, then convert a BPF filter into a Wireshark display filter
Backup if inaccessible	N06 — Wireshark User's Guide
Known limitations	Root privileges may be needed; do not capture networks you do not own
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N09 · Nmap Reference Guide

Exact title	Nmap Reference Guide
Direct URL	https://nmap.org/book/man.html
Resource type / domains	Official documentation · Modules B,G
Topic and subtopic coverage	The definitive source for scan semantics and responsible interpretation
Exact path / module / chapter / playlist	Host discovery, port scanning, service/version detection, OS detection, timing, output
Recommended difficulty / estimated time	Intermediate · 6-10 h selected
Access status	Free
Prerequisites	TCP/IP and authorization
Why selected	The definitive source for scan semantics and responsible interpretation
What to practise after	Scan only an isolated lab target with written scope; compare default, SYN and version scans
Backup if inaccessible	N10 — RFC 9293: Transmission Control Protocol
Known limitations	Many options are not appropriate for beginners or public networks
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N10 · RFC 9293: Transmission Control Protocol

Exact title	RFC 9293: Transmission Control Protocol
Direct URL	https://www.rfc-editor.org/rfc/rfc9293
Resource type / domains	Authoritative standard · Modules B
Topic and subtopic coverage	Primary standard for TCP behavior; removes myths from secondary tutorials
Exact path / module / chapter / playlist	Sections 3.1-3.10: header format, connection states, sequence numbers, handshake, retransmission, flow control
Recommended difficulty / estimated time	Advanced reference · 5-8 h selected
Access status	Free
Prerequisites	IPv4 and protocol layering
Why selected	Primary standard for TCP behavior; removes myths from secondary tutorials
What to practise after	Map every field in a real SYN/SYN-ACK/ACK capture to the RFC
Backup if inaccessible	B10 — Suricata documentation
Known limitations	RFC-level prose is dense; use after guided lessons
Classification / age status	Advanced reference · 2022, current TCP standard
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N11 · RFC 791: Internet Protocol v4

Exact title	RFC 791: Internet Protocol v4
Direct URL	https://www.rfc-editor.org/rfc/rfc791
Resource type / domains	Authoritative standard · Modules B
Topic and subtopic coverage	Primary IPv4 source for header analysis and fragmentation theory
Exact path / module / chapter / playlist	Addressing, fragmentation, header fields, TTL and protocol fields
Recommended difficulty / estimated time	Advanced reference · 3-5 h selected
Access status	Free
Prerequisites	Binary and CIDR
Why selected	Primary IPv4 source for header analysis and fragmentation theory
What to practise after	Inspect an IPv4 header in Wireshark and explain every field
Backup if inaccessible	N10 — RFC 9293: Transmission Control Protocol
Known limitations	Historic RFC; use current clarifications for security specifics
Classification / age status	Advanced reference · 1981, old but gold
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N12 · RFC 8200: Internet Protocol v6

Exact title	RFC 8200: Internet Protocol v6
Direct URL	https://www.rfc-editor.org/rfc/rfc8200
Resource type / domains	Authoritative standard · Modules B
Topic and subtopic coverage	Primary source for IPv6 differences and extension-header reasoning
Exact path / module / chapter / playlist	IPv6 header, extension headers, MTU, fragmentation and addressing model
Recommended difficulty / estimated time	Advanced reference · 4-7 h selected
Access status	Free
Prerequisites	IPv4/CIDR
Why selected	Primary source for IPv6 differences and extension-header reasoning
What to practise after	Compare an IPv6 packet capture with IPv4 and list security implications
Backup if inaccessible	N10 — RFC 9293: Transmission Control Protocol
Known limitations	Updated by later RFCs; use as baseline
Classification / age status	Advanced reference · 2017, old but gold
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N13 · Cloudflare Learning Center: What is DNS?

Exact title	Cloudflare Learning Center: What is DNS?
Direct URL	https://www.cloudflare.com/learning/dns/what-is-dns/
Resource type / domains	Educational documentation · Modules B
Topic and subtopic coverage	Clear visual explanation of the resolution chain before packet-level DNS analysis
Exact path / module / chapter / playlist	DNS resolution, recursive resolvers, root/TLD/authoritative servers, common records
Recommended difficulty / estimated time	Beginner · 1-3 h
Access status	Free
Prerequisites	Basic networking
Why selected	Clear visual explanation of the resolution chain before packet-level DNS analysis
What to practise after	Use dig or nslookup for one owned domain/example domain and trace each answer section
Backup if inaccessible	B10 — Suricata documentation
Known limitations	Vendor educational content; cross-check record behavior in packet captures
Classification / age status	Recommended · Maintained
Verification status / date	Automated checker blocked (HTTP 403); official/canonical URL; use backup if browser access fails · checked 2026-10-27

N14 · IANA Service Name and Transport Protocol Port Number Registry

Exact title	IANA Service Name and Transport Protocol Port Number Registry
Direct URL	https://www.iana.org/assignments/service-names-port-numbers/service-names-port-numbers.xhtml
Resource type / domains	Authoritative registry · Modules B,G
Topic and subtopic coverage	Avoid memorizing misleading port lists from random guides
Exact path / module / chapter / playlist	Assigned ports and service names for TCP, UDP, SCTP and DCCP
Recommended difficulty / estimated time	Reference · 1 h lookup
Access status	Free
Prerequisites	TCP/UDP fundamentals
Why selected	Avoid memorizing misleading port lists from random guides
What to practise after	Build a personal port table with service, protocol, purpose and security caveat
Backup if inaccessible	N10 — RFC 9293: Transmission Control Protocol
Known limitations	Registration is not proof a service uses that port in a particular environment
Classification / age status	Recommended · Continuously updated
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N15 · Cisco Packet Tracer course page

Exact title	Cisco Packet Tracer course page
Direct URL	https://www.netacad.com/courses/packet-tracer
Resource type / domains	Network simulator · Modules B
Topic and subtopic coverage	Low-resource alternative to physical Cisco hardware for switching and routing labs
Exact path / module / chapter / playlist	Introductory Packet Tracer course and download access
Recommended difficulty / estimated time	Beginner · 4-8 h
Access status	Free with NetAcad account availability varies by region
Prerequisites	None
Why selected	Low-resource alternative to physical Cisco hardware for switching and routing labs
What to practise after	Build the exact topologies assigned in N01/N03 and save annotated .pkt files
Backup if inaccessible	N03 — Jeremy's IT Lab CCNA Course Files
Known limitations	Registration and local availability may vary
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N16 · Subnetting Practice

Exact title	Subnetting Practice
Direct URL	https://subnettingpractice.com/
Resource type / domains	Interactive practice · Modules B
Topic and subtopic coverage	Fast, randomized drills with immediate feedback create subnetting fluency
Exact path / module / chapter / playlist	CIDR, network/broadcast/host calculations, VLSM and masks
Recommended difficulty / estimated time	Beginner to intermediate · 10-15 min daily in Month 2
Access status	Free tier; optional ads/premium features may appear
Prerequisites	Binary and IPv4 structure
Why selected	Fast, randomized drills with immediate feedback create subnetting fluency
What to practise after	Complete 20 timed questions, write error explanations for every miss
Backup if inaccessible	N17 — Practical Networking Subnetting Mastery
Known limitations	Not a conceptual textbook; use only after the lesson
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

N17 · Practical Networking Subnetting Mastery

Exact title	Practical Networking Subnetting Mastery
Direct URL	https://www.practicalnetworking.net/stand-alone/subnetting-mastery/
Resource type / domains	Course / articles · Modules B
Topic and subtopic coverage	Visual method for subnetting rather than rote tricks
Exact path / module / chapter / playlist	Seven-part subnetting series: CIDR, masks, FLSM, VLSM, shortcuts, route summarization
Recommended difficulty / estimated time	Beginner to intermediate · 6-12 h
Access status	Free articles/videos; optional paid resources
Prerequisites	Binary notation
Why selected	Visual method for subnetting rather than rote tricks
What to practise after	Draw the address block for each assigned problem and verify with ipcalc
Backup if inaccessible	N16 — Subnetting Practice
Known limitations	Course packaging and optional product prices may change
Classification / age status	Recommended · Classic, technically durable
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L01 · Linux Journey

Exact title	Linux Journey
Direct URL	https://linuxjourney.com/
Resource type / domains	Interactive text lessons · Modules C
Topic and subtopic coverage	Short, structured lessons that are easy to convert into terminal practice
Exact path / module / chapter / playlist	Grasshopper, Ostrich, Panda and Tortoise sections: command line, filesystem, processes, packages, users, permissions, networking, shell scripting
Recommended difficulty / estimated time	Beginner · 15-25 h
Access status	Free
Prerequisites	A Linux VM or WSL
Why selected	Short, structured lessons that are easy to convert into terminal practice
What to practise after	After every lesson, reproduce commands in the VM and note one failure mode
Backup if inaccessible	L02 — OverTheWire Bandit
Known limitations	Does not replace distribution documentation or systemd internals
Classification / age status	Mandatory · Maintained; older design
Verification status / date	Automated checker blocked (HTTP 403); official/canonical URL; use backup if browser access fails · checked 2026-10-27

L02 · OverTheWire Bandit

Exact title	OverTheWire Bandit
Direct URL	https://overthewire.org/wargames/bandit/
Resource type / domains	CTF / command-line lab · Modules C
Topic and subtopic coverage	Forces independent command-line problem-solving, not passive video watching
Exact path / module / chapter / playlist	Levels 0-34: SSH, files, hidden files, permissions, pipes, grep, find, cron, git, networking, shells
Recommended difficulty / estimated time	Beginner to intermediate · 10-20 h
Access status	Free
Prerequisites	SSH client and Linux basics
Why selected	Forces independent command-line problem-solving, not passive video watching
What to practise after	Keep a concept journal for each level; do not publish level passwords or copy walkthroughs verbatim
Backup if inaccessible	L01 — Linux Journey
Known limitations	Occasional service availability; some late levels rely on Unix tricks
Classification / age status	Mandatory · Classic
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L03 · Greg's Wiki BashGuide

Exact title	Greg's Wiki BashGuide
Direct URL	https://mywiki.wooledge.org/BashGuide
Resource type / domains	Free scripting guide · Modules C,D
Topic and subtopic coverage	Explains Bash correctly, including quoting and word splitting pitfalls common in security automation
Exact path / module / chapter / playlist	Commands and arguments, quoting, expansions, tests, conditionals, loops, functions, arrays, redirection
Recommended difficulty / estimated time	Intermediate · 10-15 h
Access status	Free
Prerequisites	Basic Linux CLI
Why selected	Explains Bash correctly, including quoting and word splitting pitfalls common in security automation
What to practise after	Refactor one script with shellcheck and write tests for empty strings and spaces in filenames
Backup if inaccessible	P04 — RegexOne
Known limitations	Wiki can be terse; use as a reference while coding
Classification / age status	Recommended · Classic, technically durable
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L04 · Linux man-pages project

Exact title	Linux man-pages project
Direct URL	https://man7.org/linux/man-pages/
Resource type / domains	Official reference · Modules C
Topic and subtopic coverage	Primary local/online reference for exact command and kernel API behavior
Exact path / module / chapter / playlist	Sections 1, 2, 3, 4, 5, 7 and 8; start with ls, chmod, ps, mount, namespaces, capabilities, proc, syscalls
Recommended difficulty / estimated time	Intermediate · Reference use
Access status	Free
Prerequisites	Linux basics
Why selected	Primary local/online reference for exact command and kernel API behavior
What to practise after	Replace memorized commands with a man-page lookup; cite section numbers in notes
Backup if inaccessible	L01 — Linux Journey
Known limitations	Distribution versions may differ; compare with local man pages
Classification / age status	Advanced reference · Continuously updated
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L05 · systemd.unit manual

Exact title	systemd.unit manual
Direct URL	https://www.freedesktop.org/software/systemd/man/latest/systemd.unit.html
Resource type / domains	Official reference · Modules C
Topic and subtopic coverage	Authoritative service configuration and troubleshooting reference
Exact path / module / chapter / playlist	Unit file structure, sections, dependencies, targets and drop-ins
Recommended difficulty / estimated time	Intermediate · 2-4 h
Access status	Free
Prerequisites	Linux services
Why selected	Authoritative service configuration and troubleshooting reference
What to practise after	Create a benign timer/service pair, inspect journal logs, then disable and remove it
Backup if inaccessible	L04 — Linux man-pages project
Known limitations	Not all distributions use identical defaults
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L06 · The Linux Foundation Introduction to Linux

Exact title	The Linux Foundation Introduction to Linux
Direct URL	https://www.edx.org/learn/linux/the-linux-foundation-introduction-to-linux
Resource type / domains	MOOC · Modules C
Topic and subtopic coverage	University-like sequence by the organization behind Linux training
Exact path / module / chapter / playlist	Filesystem, command line, processes, packages, users, permissions, shells, scripting
Recommended difficulty / estimated time	Beginner · 40-60 h; audit selected modules
Access status	Free audit may be time-limited; verified certificate paid
Prerequisites	None
Why selected	University-like sequence by the organization behind Linux training
What to practise after	Use selected chapters to fill Linux Journey gaps; do all terminal exercises locally
Backup if inaccessible	L01 — Linux Journey
Known limitations	edX audit terms and course versions vary
Classification / age status	Optional enrichment · Maintained with periodic updates
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L07 · Microsoft Windows Server documentation

Exact title	Microsoft Windows Server documentation
Direct URL	https://learn.microsoft.com/en-us/windows-server/
Resource type / domains	Official documentation · Modules C
Topic and subtopic coverage	Primary source for Windows Server and Active Directory concepts
Exact path / module / chapter / playlist	Installation, administration, identity, networking, security, PowerShell, failover and containers
Recommended difficulty / estimated time	Beginner to intermediate · Reference use
Access status	Free
Prerequisites	Windows client basics
Why selected	Primary source for Windows Server and Active Directory concepts
What to practise after	Create a small virtual domain only if your hardware allows; otherwise diagram it and use a local Windows event lab
Backup if inaccessible	L12 — Active Directory Domain Services overview
Known limitations	Broad documentation; use the daily plan's exact pages
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L08 · PowerShell documentation

Exact title	PowerShell documentation
Direct URL	https://learn.microsoft.com/en-us/powershell/
Resource type / domains	Official documentation · Modules C,D
Topic and subtopic coverage	Primary source for current PowerShell syntax and security controls
Exact path / module / chapter / playlist	Getting started, scripting, modules, remoting, security and cmdlet reference
Recommended difficulty / estimated time	Beginner to intermediate · 8-15 h
Access status	Free
Prerequisites	Windows command line
Why selected	Primary source for current PowerShell syntax and security controls
What to practise after	Write a benign inventory script and export structured JSON/CSV
Backup if inaccessible	L03 — Greg's Wiki BashGuide
Known limitations	Windows PowerShell 5.1 and PowerShell 7 differ; check edition
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L09 · Windows Event Log documentation

Exact title	Windows Event Log documentation
Direct URL	https://learn.microsoft.com/en-us/windows/win32/wes/windows-event-log
Resource type / domains	Official reference · Modules C,H
Topic and subtopic coverage	Definitive event-log architecture reference for blue-team work
Exact path / module / chapter / playlist	Event logs, channels, records, queries, subscriptions and security
Recommended difficulty / estimated time	Intermediate · 3-6 h
Access status	Free
Prerequisites	Windows administration
Why selected	Definitive event-log architecture reference for blue-team work
What to practise after	Export a sample EVTX, identify Channel, EventID, Provider and RecordId
Backup if inaccessible	L11 — Sysmon documentation and download
Known limitations	Some APIs are developer-oriented; use Event Viewer for first practice
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L10 · Microsoft Sysinternals

Exact title	Microsoft Sysinternals
Direct URL	https://learn.microsoft.com/en-us/sysinternals/
Resource type / domains	Official tools/docs · Modules C,H
Topic and subtopic coverage	Industry-standard, free Windows internals tools with official documentation
Exact path / module / chapter / playlist	Process Explorer, Process Monitor, Autoruns, Sysmon and PsTools
Recommended difficulty / estimated time	Beginner to intermediate · 5-10 h
Access status	Free
Prerequisites	Windows fundamentals
Why selected	Industry-standard, free Windows internals tools with official documentation
What to practise after	Investigate a benign process launch and document process tree, loaded modules, handles and autoruns
Backup if inaccessible	L08 — PowerShell documentation
Known limitations	Tools change; use current downloads and hashes
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L11 · Sysmon documentation and download

Exact title	Sysmon documentation and download
Direct URL	https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon
Resource type / domains	Official tool/doc · Modules H
Topic and subtopic coverage	Adds high-value process, network and file events for detection engineering in an authorized lab
Exact path / module / chapter / playlist	Installation, configuration, event IDs and schema
Recommended difficulty / estimated time	Intermediate · 3-6 h
Access status	Free
Prerequisites	Windows event logs
Why selected	Adds high-value process, network and file events for detection engineering in an authorized lab
What to practise after	Install Sysmon in a disposable VM and generate benign process/network telemetry
Backup if inaccessible	L09 — Windows Event Log documentation
Known limitations	Do not deploy on production systems without organizational approval
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L12 · Active Directory Domain Services overview

Exact title	Active Directory Domain Services overview
Direct URL	https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/get-started/virtual-dc/active-directory-domain-services-overview
Resource type / domains	Official documentation · Modules C,E
Topic and subtopic coverage	Authoritative identity architecture reference before AD security topics
Exact path / module / chapter / playlist	Forest, domain, OU, users, groups, DNS, trusts and authentication
Recommended difficulty / estimated time	Beginner to intermediate · 4-8 h
Access status	Free
Prerequisites	Windows Server basics
Why selected	Authoritative identity architecture reference before AD security topics
What to practise after	Create a diagram of a fictional domain and label every identity boundary
Backup if inaccessible	L07 — Microsoft Windows Server documentation
Known limitations	UI and server version differ across environments
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L13 · Windows security baselines

Exact title	Windows security baselines
Direct URL	https://learn.microsoft.com/en-us/windows/security/operating-system-security/device-management/windows-security-configuration-framework/windows-security-baselines
Resource type / domains	Official configuration guidance · Modules C,E
Topic and subtopic coverage	Real enterprise hardening baselines rather than ad hoc tweaks
Exact path / module / chapter / playlist	Microsoft security baseline packages, policy analyzer, deployment guidance and versioning
Recommended difficulty / estimated time	Intermediate · 4-8 h
Access status	Free
Prerequisites	Group Policy and Windows administration
Why selected	Real enterprise hardening baselines rather than ad hoc tweaks
What to practise after	Compare one baseline setting against a clean VM, document compatibility impact
Backup if inaccessible	L14 — CIS Benchmarks
Known limitations	Do not apply a baseline blindly to production or your primary computer
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L14 · CIS Benchmarks

Exact title	CIS Benchmarks
Direct URL	https://www.cisecurity.org/cis-benchmarks
Resource type / domains	Consensus benchmarks · Modules C,E,I,J
Topic and subtopic coverage	Detailed consensus controls with rationale and audit/remediation steps
Exact path / module / chapter / playlist	Benchmarks by OS, cloud, database, container and network platform
Recommended difficulty / estimated time	Intermediate · Reference use
Access status	Free PDFs may require registration; some tools paid
Prerequisites	Hardening basics
Why selected	Detailed consensus controls with rationale and audit/remediation steps
What to practise after	Select five controls for Ubuntu or Windows, test in a snapshot, record before/after evidence
Backup if inaccessible	L13 — Windows security baselines
Known limitations	Registration and profile versions; not all benchmarks fit student hardware
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

L15 · pwn.college Dojo index

Exact title	pwn.college Dojo index
Direct URL	https://pwn.college/dojos
Resource type / domains	Hands-on challenges · Modules A,C,K
Topic and subtopic coverage	University-grade challenge sequence with a browser workspace; excellent low-hardware option
Exact path / module / chapter / playlist	Linux Luminarium, Computing 101, Intro to Cybersecurity, Program Security, System Security, Software Exploitation
Recommended difficulty / estimated time	Beginner to advanced · Varies by dojo
Access status	Free
Prerequisites	Command line and programming
Why selected	University-grade challenge sequence with a browser workspace; excellent low-hardware option
What to practise after	Use Computing 101 and Linux Fundamentals in Months 1-4; defer exploitation dojos until prerequisites are met
Backup if inaccessible	L01 — Linux Journey
Known limitations	Dojo names and course instances change; use current Core Material list
Classification / age status	Recommended · Maintained 2026
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

P01 · Python for Everybody

Exact title	Python for Everybody
Direct URL	https://www.py4e.com/
Resource type / domains	University course · Modules D
Topic and subtopic coverage	Very gentle programming sequence with abundant exercises
Exact path / module / chapter / playlist	Chapters 1-16: variables, conditionals, functions, loops, strings, files, lists, dictionaries, tuples, regex, networking, databases, automation
Recommended difficulty / estimated time	Beginner · 30-50 h; selected chapters in Month 4
Access status	Free site; Coursera specialization freemium
Prerequisites	None
Why selected	Very gentle programming sequence with abundant exercises
What to practise after	Complete selected chapter exercises and adapt one to parse a security log
Backup if inaccessible	F08 — Python 3 Official Tutorial
Known limitations	Some videos are older; Python 3 syntax remains valid with minor updates
Classification / age status	Recommended · Old but gold
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

P02 · Learn Python - Full Course for Beginners (freeCodeCamp)

Exact title	Learn Python - Full Course for Beginners (freeCodeCamp)
Direct URL	https://www.youtube.com/watch?v=eWRfhZUzrAc
Resource type / domains	Single full-length video course · Modules D
Topic and subtopic coverage	One coherent video session avoids jumping among dozens of tutorials
Exact path / module / chapter / playlist	Chapters shown in video; variables, strings, lists, tuples, dictionaries, conditionals, loops, functions, classes and projects
Recommended difficulty / estimated time	Beginner · 4.5 h
Access status	Free
Prerequisites	None
Why selected	One coherent video session avoids jumping among dozens of tutorials
What to practise after	Watch a 25-minute segment, code all examples locally, then write a variation
Backup if inaccessible	F08 — Python 3 Official Tutorial
Known limitations	Single video chapter times can change with re-uploads; use YouTube's displayed chapters
Classification / age status	Recommended · 2022
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

P03 · Automate the Boring Stuff with Python

Exact title	Automate the Boring Stuff with Python
Direct URL	https://automatetheboringstuff.com/
Resource type / domains	Free online book · Modules D
Topic and subtopic coverage	Practical automation tasks map directly to defensive utility building
Exact path / module / chapter / playlist	Chapters 1-18; focus on files, regex, CSV/JSON, web scraping, requests, Excel, email and input validation
Recommended difficulty / estimated time	Beginner to intermediate · 20-40 h selected
Access status	Free online; paid print/course optional
Prerequisites	Basic Python
Why selected	Practical automation tasks map directly to defensive utility building
What to practise after	Build the Month 4 log parser and audit utility from the book's patterns
Backup if inaccessible	F08 — Python 3 Official Tutorial
Known limitations	Some third-party APIs or screenshots are dated; use current library docs
Classification / age status	Recommended · Classic, current Python edition
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

P04 · RegexOne

Exact title	RegexOne
Direct URL	https://regexone.com/
Resource type / domains	Interactive lessons · Modules D
Topic and subtopic coverage	Short, interactive regex practice with immediate feedback
Exact path / module / chapter / playlist	Lessons 1-15: literals, wildcards, character classes, anchors, groups, quantifiers, backreferences and practical examples
Recommended difficulty / estimated time	Beginner · 3-6 h
Access status	Free
Prerequisites	None
Why selected	Short, interactive regex practice with immediate feedback
What to practise after	Write regexes for three synthetic log formats and document false positives
Backup if inaccessible	F08 — Python 3 Official Tutorial
Known limitations	Regex dialects differ; verify in Python re docs
Classification / age status	Recommended · Classic, technically durable
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

P05 · pytest documentation

Exact title	pytest documentation
Direct URL	https://docs.pytest.org/en/stable/
Resource type / domains	Official documentation · Modules D
Topic and subtopic coverage	Reliable testing habits for security scripts and parsers
Exact path / module / chapter / playlist	Getting started, fixtures, parametrization, assertions, marking, plugins and CI examples
Recommended difficulty / estimated time	Intermediate · 5-10 h
Access status	Free
Prerequisites	Python functions and modules
Why selected	Reliable testing habits for security scripts and parsers
What to practise after	Write unit tests for normal, malformed and adversarial inputs to the log parser
Backup if inaccessible	F08 — Python 3 Official Tutorial
Known limitations	Plugin compatibility varies
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

P06 · Requests: HTTP for Humans

Exact title	Requests: HTTP for Humans
Direct URL	https://requests.readthedocs.io/en/latest/
Resource type / domains	Official library docs · Modules D,F
Topic and subtopic coverage	Safe, documented HTTP client usage with timeouts and TLS verification
Exact path / module / chapter / playlist	Quickstart, advanced usage, authentication, sessions, SSL verification, timeouts and streaming
Recommended difficulty / estimated time	Beginner to intermediate · 3-6 h
Access status	Free
Prerequisites	Python and HTTP basics
Why selected	Safe, documented HTTP client usage with timeouts and TLS verification
What to practise after	Query a mock API or local test server; record status codes, headers, errors and retries
Backup if inaccessible	F10 — MDN HTTP Overview
Known limitations	Never disable certificate verification except in a controlled local TLS lab
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

P07 · SQLite documentation

Exact title	SQLite documentation
Direct URL	https://www.sqlite.org/docs.html
Resource type / domains	Official documentation · Modules A,D
Topic and subtopic coverage	Lightweight local database for capstone and log-analysis projects
Exact path / module / chapter / playlist	SQL syntax, datatypes, indexes, transactions, command-line shell and security considerations
Recommended difficulty / estimated time	Beginner to intermediate · Reference use
Access status	Free
Prerequisites	SQLBolt basics
Why selected	Lightweight local database for capstone and log-analysis projects
What to practise after	Create a normalized schema for synthetic events and write parameterized queries
Backup if inaccessible	F11 — SQLBolt
Known limitations	SQLite is not a substitute for learning PostgreSQL/MySQL administration
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

P08 · OWASP Secure Coding Practices Quick Reference Guide

Exact title	OWASP Secure Coding Practices Quick Reference Guide
Direct URL	https://owasp.org/www-project-secure-coding-practices-quick-reference-guide/
Resource type / domains	Checklist/reference · Modules D,F
Topic and subtopic coverage	Maps secure implementation decisions to concrete verification checks
Exact path / module / chapter / playlist	Input validation, output encoding, authentication, session management, access control, cryptography, error handling, data protection, communications and system configuration
Recommended difficulty / estimated time	Intermediate · 4-8 h
Access status	Free
Prerequisites	Programming and web basics
Why selected	Maps secure implementation decisions to concrete verification checks
What to practise after	Apply ten relevant checklist items to your own secure API project
Backup if inaccessible	W01 — PortSwigger Web Security Academy
Known limitations	Checklist, not a tutorial or complete standard
Classification / age status	Mandatory · Community-maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C01 · Professor Messer SY0-701 Security+ Course Index

Exact title	Professor Messer SY0-701 Security+ Course Index
Direct URL	https://www.professormesser.com/security-plus/sy0-701/sy0-701-video/sy0-701-comptia-security-plus-course/
Resource type / domains	Course index · Modules E,J
Topic and subtopic coverage	Objective-indexed, concise, fully free core security theory coverage
Exact path / module / chapter / playlist	121 videos grouped by SY0-701 objectives 1.1-5.6
Recommended difficulty / estimated time	Beginner to intermediate · 15 h 11 m total
Access status	Free videos; paid notes/practice exams optional
Prerequisites	None
Why selected	Objective-indexed, concise, fully free core security theory coverage
What to practise after	Use one video per assigned objective to supplement CS50 and hands-on labs
Backup if inaccessible	C02 — Professor Messer SY0-701 Security+ official playlist
Known limitations	SY0-701 is V7; V8/SY0-801 launches Nov 17 2026, so use current objectives if certifying later
Classification / age status	Mandatory · 2023-2024, exam-relevant until stated retirement
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C02 · Professor Messer SY0-701 Security+ official playlist

Exact title	Professor Messer SY0-701 Security+ official playlist
Direct URL	https://www.youtube.com/playlist?list=PLG49S3nxzAnl4QDVqK-hOnoqcSKEIDDuv
Resource type / domains	YouTube playlist · Modules E,J
Topic and subtopic coverage	Exact playlist rather than channel home; short videos suit retrieval practice
Exact path / module / chapter / playlist	Official 121-video SY0-701 playlist; match objective code in video title
Recommended difficulty / estimated time	Beginner to intermediate · 15 h 11 m total
Access status	Free
Prerequisites	None
Why selected	Exact playlist rather than channel home; short videos suit retrieval practice
What to practise after	Watch only the objective in the daily plan; do not binge entire playlists
Backup if inaccessible	C01 — Professor Messer SY0-701 Security+ Course Index
Known limitations	V8 content will differ after Nov 2026
Classification / age status	Recommended · 2023-2024
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C03 · ISC2 Certified in Cybersecurity Exam Outline

Exact title	ISC2 Certified in Cybersecurity Exam Outline
Direct URL	https://www.isc2.org/certifications/cc/cc-certification-exam-outline
Resource type / domains	Official exam outline · Modules E,J
Topic and subtopic coverage	Current official scope for the free/low-cost entry-level certification option
Exact path / module / chapter / playlist	Five domains; outline effective September 1 2026; AI concepts integrated across domains
Recommended difficulty / estimated time	Beginner · 2-4 h
Access status	Free
Prerequisites	None
Why selected	Current official scope for the free/low-cost entry-level certification option
What to practise after	Map each Month 5 topic to an outline domain and create flashcards
Backup if inaccessible	C04 — ISC2 CC self-study resources
Known limitations	Exam domains change; download current outline before paying
Classification / age status	Recommended · Effective 2026-09-01
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C04 · ISC2 CC self-study resources

Exact title	ISC2 CC self-study resources
Direct URL	https://www.isc2.org/certifications/cc/cc-self-study-resources
Resource type / domains	Official study resources · Modules E,J
Topic and subtopic coverage	Verified official free preparation options for CC
Exact path / module / chapter / playlist	Exam outline, flash cards, official training links and community resources
Recommended difficulty / estimated time	Beginner · 10-20 h
Access status	Free resources; official training paid
Prerequisites	None
Why selected	Verified official free preparation options for CC
What to practise after	Use flashcards only after practical exercises; write original examples for each term
Backup if inaccessible	C03 — ISC2 Certified in Cybersecurity Exam Outline
Known limitations	Marketing pages may change; exam outline controls
Classification / age status	Optional enrichment · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C05 · ISC2 exam pricing

Exact title	ISC2 exam pricing
Direct URL	https://www.isc2.org/register-for-exam/isc2-exam-pricing
Resource type / domains	Official pricing · Modules J
Topic and subtopic coverage	Provides verified current price instead of relying on outdated blogs
Exact path / module / chapter / playlist	CC exam price by region; rescheduling and cancellation fees
Recommended difficulty / estimated time	Reference · 10 min
Access status	Free to view; exam paid
Prerequisites	None
Why selected	Provides verified current price instead of relying on outdated blogs
What to practise after	Check Pakistan-inclusive region at checkout before budgeting
Backup if inaccessible	C03 — ISC2 Certified in Cybersecurity Exam Outline
Known limitations	Local taxes and currencies vary; price may change
Classification / age status	Recommended · Checked 2026-10-27; CC listed US$199 in listed regions
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C06 · NIST Cybersecurity Framework 2.0

Exact title	NIST Cybersecurity Framework 2.0
Direct URL	https://www.nist.gov/cyberframework
Resource type / domains	Framework · Modules E,J
Topic and subtopic coverage	Current, vendor-neutral framework for organizing a security program
Exact path / module / chapter / playlist	Govern, Identify, Protect, Detect, Respond, Recover; implementation examples and quick-start guides
Recommended difficulty / estimated time	Beginner to intermediate · 6-12 h
Access status	Free
Prerequisites	Core security vocabulary
Why selected	Current, vendor-neutral framework for organizing a security program
What to practise after	Build a mini CSF profile for your home lab and map evidence to each function
Backup if inaccessible	C21 — CIS Controls Version 8
Known limitations	Framework is not law and not a certification
Classification / age status	Mandatory · CSF 2.0 current
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C07 · NIST SP 800-61 Revision 3

Exact title	NIST SP 800-61 Revision 3
Direct URL	https://csrc.nist.gov/pubs/sp/800/61/r3/final
Resource type / domains	Official guidance · Modules E,H,J
Topic and subtopic coverage	Current authoritative IR guidance connecting technical work to governance
Exact path / module / chapter / playlist	Incident response life cycle, CSF alignment, preparation, detection, analysis, containment, eradication, recovery and improvement
Recommended difficulty / estimated time	Intermediate · 5-8 h
Access status	Free
Prerequisites	Logging and networking basics
Why selected	Current authoritative IR guidance connecting technical work to governance
What to practise after	Write an IR plan and runbook for one simulated lab incident
Backup if inaccessible	C06 — NIST Cybersecurity Framework 2.0
Known limitations	Guidance is advisory for US federal systems; other jurisdictions may differ
Classification / age status	Mandatory · 2025
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C08 · NIST SP 800-63B

Exact title	NIST SP 800-63B
Direct URL	https://pages.nist.gov/800-63-4/sp800-63b.html
Resource type / domains	Official guidance · Modules E,F
Topic and subtopic coverage	Current digital identity guidance used in modern authentication decisions
Exact path / module / chapter / playlist	Authentication and authenticator management; passwords, MFA, phishing resistance, lifecycle and assurance
Recommended difficulty / estimated time	Intermediate · 5-10 selected pages
Access status	Free
Prerequisites	Identity basics
Why selected	Current digital identity guidance used in modern authentication decisions
What to practise after	Review a sample login design and list five control gaps
Backup if inaccessible	C09 — Crypto 101
Known limitations	Draft/revision status and version changes; use the page's current edition marker
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C09 · Crypto 101

Exact title	Crypto 101
Direct URL	https://www.crypto101.io/
Resource type / domains	Free course/book · Modules E
Topic and subtopic coverage	Concise, implementation-aware cryptography introduction with exercises
Exact path / module / chapter / playlist	XOR, block ciphers, stream ciphers, hash functions, MACs, key exchange, public-key encryption, signatures and TLS concepts
Recommended difficulty / estimated time	Intermediate · 10-20 h
Access status	Free
Prerequisites	Python and basic algebra
Why selected	Concise, implementation-aware cryptography introduction with exercises
What to practise after	Implement only toy demonstrations in Python; never build production crypto
Backup if inaccessible	C10 — Stanford Cryptography I
Known limitations	Course is educational, not a substitute for vetted libraries or proofs
Classification / age status	Recommended · Classic
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C10 · Stanford Cryptography I

Exact title	Stanford Cryptography I
Direct URL	https://www.coursera.org/learn/crypto
Resource type / domains	University course · Modules E
Topic and subtopic coverage	Rigorous cryptography taught by a leading researcher; excellent second exposure after CS50
Exact path / module / chapter / playlist	Week 1-6 lecture list: stream ciphers, block ciphers, message integrity, authenticated encryption, key exchange, public-key encryption and signatures
Recommended difficulty / estimated time	Intermediate to advanced · 20-40 h
Access status	Free audit/lectures may be available; verified certificate and graded homework paid
Prerequisites	Discrete math and programming
Why selected	Rigorous cryptography taught by a leading researcher; excellent second exposure after CS50
What to practise after	Watch one selected lecture, then explain the security goal and attack model in a notebook
Backup if inaccessible	C09 — Crypto 101
Known limitations	Older course; primitives remain important but verify current protocol guidance
Classification / age status	Optional enrichment · Old but gold
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C11 · A Graduate Course in Applied Cryptography

Exact title	A Graduate Course in Applied Cryptography
Direct URL	https://toc.cryptobook.us/
Resource type / domains	Free textbook · Modules E
Topic and subtopic coverage	Free modern cryptography reference when you need exact definitions and proof intuition
Exact path / module / chapter / playlist	Symmetric ciphers, MACs, hash functions, authenticated encryption, public-key crypto, digital signatures and proofs
Recommended difficulty / estimated time	Advanced · Reference use
Access status	Free
Prerequisites	Discrete math, probability and algorithms
Why selected	Free modern cryptography reference when you need exact definitions and proof intuition
What to practise after	Look up one theorem or construction from the lecture and write a plain-English summary
Backup if inaccessible	C09 — Crypto 101
Known limitations	Graduate-level; not a Month 5 mandatory read
Classification / age status	Advanced reference · Maintained draft
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C12 · Let's Encrypt documentation

Exact title	Let's Encrypt documentation
Direct URL	https://letsencrypt.org/docs/
Resource type / domains	Official documentation · Modules E,F
Topic and subtopic coverage	Current practical documentation for certificate issuance and lifecycle
Exact path / module / chapter / playlist	Certificates, ACME, chain of trust, validation, renewal, certificate transparency and best practices
Recommended difficulty / estimated time	Beginner to intermediate · 3-6 h
Access status	Free
Prerequisites	DNS and HTTP
Why selected	Current practical documentation for certificate issuance and lifecycle
What to practise after	Inspect a certificate chain in a browser and explain each field; do not deploy on public sites without authorization
Backup if inaccessible	C13 — RFC 8446: TLS 1.3
Known limitations	Vendor-specific ACME details; general PKI principles still apply
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C13 · RFC 8446: TLS 1.3

Exact title	RFC 8446: TLS 1.3
Direct URL	https://www.rfc-editor.org/rfc/rfc8446
Resource type / domains	Authoritative standard · Modules E,F
Topic and subtopic coverage	Primary TLS 1.3 reference for advanced protocol understanding
Exact path / module / chapter / playlist	Handshake, key schedule, record layer, certificates, resumption, 0-RTT and security considerations
Recommended difficulty / estimated time	Advanced · 8-15 selected pages
Access status	Free
Prerequisites	Cryptography and TCP/IP
Why selected	Primary TLS 1.3 reference for advanced protocol understanding
What to practise after	Annotate a TLS 1.3 handshake capture and identify the client/server hello extensions
Backup if inaccessible	C12 — Let's Encrypt documentation
Known limitations	Dense standard; use simplified material first
Classification / age status	Advanced reference · 2018, current baseline
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C14 · OWASP Top 10

Exact title	OWASP Top 10
Direct URL	https://owasp.org/www-project-top-ten/
Resource type / domains	Reference · Modules F
Topic and subtopic coverage	Industry-recognized risk categories; best as an index, not a complete course
Exact path / module / chapter / playlist	A01-A10 web application security risks, mappings and prevention guidance
Recommended difficulty / estimated time	Beginner to intermediate · 4-8 h
Access status	Free
Prerequisites	HTTP and web app basics
Why selected	Industry-recognized risk categories; best as an index, not a complete course
What to practise after	For each category, point to a PortSwigger lab and a secure control in your own project
Backup if inaccessible	W01 — PortSwigger Web Security Academy
Known limitations	2021 edition; check current release before citing year in interviews
Classification / age status	Mandatory · Community-maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C15 · OWASP Application Security Verification Standard

Exact title	OWASP Application Security Verification Standard
Direct URL	https://owasp.org/www-project-application-security-verification-standard/
Resource type / domains	Standard/reference · Modules F
Topic and subtopic coverage	Concrete verification requirements for secure engineering rather than vague advice
Exact path / module / chapter / playlist	V1 architecture, V2 authentication, V3 session, V4 access control, V5 validation, V6 cryptography, V7 errors/logging, V8 data protection, V9 communications, V10 malicious code and more
Recommended difficulty / estimated time	Intermediate · 6-12 h selected
Access status	Free
Prerequisites	Web security fundamentals
Why selected	Concrete verification requirements for secure engineering rather than vague advice
What to practise after	Select Level 1 requirements and write pass/fail tests for your secure API
Backup if inaccessible	P08 — OWASP Secure Coding Practices Quick Reference Guide
Known limitations	Formal standard; not all levels apply to student projects
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C16 · OWASP API Security Top 10

Exact title	OWASP API Security Top 10
Direct URL	https://owasp.org/API-Security/editions/2023/en/0x11-t10/
Resource type / domains	Reference · Modules F
Topic and subtopic coverage	API-specific risk taxonomy missing from general web courses
Exact path / module / chapter / playlist	API1-API10: BOLA, authentication, object property authorization, resources, BFLA, business flows, SSRF, misconfiguration, inventory and unsafe APIs
Recommended difficulty / estimated time	Intermediate · 4-8 h
Access status	Free
Prerequisites	HTTP, APIs and authorization
Why selected	API-specific risk taxonomy missing from general web courses
What to practise after	Review an OpenAPI spec or sample API and assign each risk a test case
Backup if inaccessible	C14 — OWASP Top 10
Known limitations	2023 edition; verify current release later
Classification / age status	Mandatory · 2023
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C17 · OWASP Threat Modeling community page

Exact title	OWASP Threat Modeling community page
Direct URL	https://owasp.org/www-community/Threat_Modeling
Resource type / domains	Reference/process · Modules E,F,K
Topic and subtopic coverage	Short, practical entry point before more formal methodologies
Exact path / module / chapter / playlist	Threat modeling process, assets, trust boundaries, data flows, threats and mitigations
Recommended difficulty / estimated time	Intermediate · 3-6 h
Access status	Free
Prerequisites	System design basics
Why selected	Short, practical entry point before more formal methodologies
What to practise after	Draw a DFD for your capstone and enumerate STRIDE threats per element
Backup if inaccessible	C18 — MITRE ATT&CK
Known limitations	Community page is an overview, not a complete methodology
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C18 · MITRE ATT&CK

Exact title	MITRE ATT&CK
Direct URL	https://attack.mitre.org/
Resource type / domains	Knowledge base · Modules E,H,K
Topic and subtopic coverage	Common language for adversary behavior and detection coverage
Exact path / module / chapter / playlist	Tactics, techniques, sub-techniques, procedures, mitigations, detections, groups and software
Recommended difficulty / estimated time	Intermediate · Reference use
Access status	Free
Prerequisites	Logs, OS and networking
Why selected	Common language for adversary behavior and detection coverage
What to practise after	Pick one technique, read its procedure examples and map synthetic telemetry to data sources
Backup if inaccessible	C19 — MITRE D3FEND
Known limitations	Not a maturity model or proof of detection coverage
Classification / age status	Mandatory · Continuously updated
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C19 · MITRE D3FEND

Exact title	MITRE D3FEND
Direct URL	https://d3fend.mitre.org/
Resource type / domains	Knowledge base · Modules E,H,K
Topic and subtopic coverage	Balances attacker-centric thinking with concrete defensive engineering options
Exact path / module / chapter / playlist	Defensive countermeasure taxonomy mapped to ATT&CK techniques
Recommended difficulty / estimated time	Intermediate to advanced · Reference use
Access status	Free
Prerequisites	ATT&CK basics
Why selected	Balances attacker-centric thinking with concrete defensive engineering options
What to practise after	For one ATT&CK technique, propose two D3FEND countermeasures and evidence needed
Backup if inaccessible	C18 — MITRE ATT&CK
Known limitations	Research project; not all entries have turnkey implementations
Classification / age status	Optional enrichment · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C20 · NIST NICE Workforce Framework

Exact title	NIST NICE Workforce Framework
Direct URL	https://www.nist.gov/itl/applied-cybersecurity/nice
Resource type / domains	Framework · Modules J
Topic and subtopic coverage	Useful international reference for analyzing job descriptions and planning skill evidence
Exact path / module / chapter / playlist	Work roles, knowledge, skills, abilities and tasks; NICE Framework resources
Recommended difficulty / estimated time	Beginner to intermediate · 3-8 h
Access status	Free
Prerequisites	None
Why selected	Useful international reference for analyzing job descriptions and planning skill evidence
What to practise after	Map your capstone evidence to three NICE work-role task statements
Backup if inaccessible	C06 — NIST Cybersecurity Framework 2.0
Known limitations	US-origin framework; employers use many different vocabularies
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C21 · CIS Controls Version 8

Exact title	CIS Controls Version 8
Direct URL	https://www.cisecurity.org/controls/v8
Resource type / domains	Best-practice controls · Modules E,J
Topic and subtopic coverage	Practical prioritized baseline for small environments and labs
Exact path / module / chapter / playlist	18 safeguard categories, implementation groups and asset/control mappings
Recommended difficulty / estimated time	Beginner to intermediate · 6-10 h
Access status	Free overview; some tools paid
Prerequisites	Basic IT asset concepts
Why selected	Practical prioritized baseline for small environments and labs
What to practise after	Create a before/after implementation-group 1 checklist for your VMs
Backup if inaccessible	C06 — NIST Cybersecurity Framework 2.0
Known limitations	Not regulation; legal obligations depend on jurisdiction
Classification / age status	Recommended · Version 8
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C22 · NIST SP 800-53 Revision 5

Exact title	NIST SP 800-53 Revision 5
Direct URL	https://csrc.nist.gov/pubs/sp/800/53/r5/upd1/final
Resource type / domains	Official controls catalog · Modules J
Topic and subtopic coverage	Authoritative control catalog for understanding enterprise security requirements
Exact path / module / chapter / playlist	Control families and baselines for information systems and organizations
Recommended difficulty / estimated time	Advanced reference · Reference use
Access status	Free
Prerequisites	Security architecture and GRC vocabulary
Why selected	Authoritative control catalog for understanding enterprise security requirements
What to practise after	Look up AC, AU, CM, IR and SI controls relevant to your lab; do not try to read linearly
Backup if inaccessible	C06 — NIST Cybersecurity Framework 2.0
Known limitations	Large catalog; use search and mappings
Classification / age status	Advanced reference · Current revision
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C23 · ISO/IEC 27001 overview

Exact title	ISO/IEC 27001 overview
Direct URL	https://www.iso.org/standard/27001
Resource type / domains	Standards page · Modules J
Topic and subtopic coverage	Distinguishes a management-system standard from technical best practices
Exact path / module / chapter / playlist	Requirements for an information security management system; standard text is normally paid
Recommended difficulty / estimated time	Beginner to intermediate · 2-4 h overview
Access status	Page free; standard document paid
Prerequisites	Basic governance vocabulary
Why selected	Distinguishes a management-system standard from technical best practices
What to practise after	Write a fictional ISMS scope, asset inventory and risk treatment plan
Backup if inaccessible	C06 — NIST Cybersecurity Framework 2.0
Known limitations	Full standard is not freely distributable; do not rely on unauthorized copies
Classification / age status	Optional enrichment · Current ISO catalog entry
Verification status / date	Automated checker blocked (HTTP 403); official/canonical URL; use backup if browser access fails · checked 2026-10-27

C24 · GDPR full text (EUR-Lex)

Exact title	GDPR full text (EUR-Lex)
Direct URL	https://eur-lex.europa.eu/eli/reg/2016/679/oj
Resource type / domains	Regulation/legal text · Modules J
Topic and subtopic coverage	Primary legal text for privacy-by-design basics
Exact path / module / chapter / playlist	Principles, lawful basis, data subject rights, controller/processor duties, breach notification, transfers and accountability
Recommended difficulty / estimated time	Intermediate · Selected articles 4-6 h
Access status	Free
Prerequisites	Privacy vocabulary
Why selected	Primary legal text for privacy-by-design basics
What to practise after	Extract ten technical privacy requirements from Articles 5, 25, 32, 33 and 35
Backup if inaccessible	C25 — NIST Privacy Framework
Known limitations	EU regulation, not global law; seek legal advice for actual obligations
Classification / age status	Recommended · Official consolidated text
Verification status / date	Accepted/asynchronous response (HTTP 202); official legal text; use backup if browser fails · checked 2026-10-27

C25 · NIST Privacy Framework

Exact title	NIST Privacy Framework
Direct URL	https://www.nist.gov/privacy-framework
Resource type / domains	Framework · Modules J
Topic and subtopic coverage	Complements CSF with privacy risk management without presenting itself as law
Exact path / module / chapter / playlist	Identify-P, Govern-P, Control-P, Communicate-P and Protect-P functions
Recommended difficulty / estimated time	Intermediate · 3-6 h
Access status	Free
Prerequisites	Basic data governance
Why selected	Complements CSF with privacy risk management without presenting itself as law
What to practise after	Add privacy controls to the capstone data-flow diagram
Backup if inaccessible	C06 — NIST Cybersecurity Framework 2.0
Known limitations	Voluntary framework, not legal compliance proof
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C26 · NIST SP 800-30 Revision 1

Exact title	NIST SP 800-30 Revision 1
Direct URL	https://csrc.nist.gov/pubs/sp/800/30/r1/final
Resource type / domains	Official risk guidance · Modules J
Topic and subtopic coverage	Methodical risk assessment vocabulary for project reports
Exact path / module / chapter / playlist	Threat sources, vulnerabilities, likelihood, impact, risk assessment process and risk registers
Recommended difficulty / estimated time	Intermediate · 5-8 h
Access status	Free
Prerequisites	Security fundamentals
Why selected	Methodical risk assessment vocabulary for project reports
What to practise after	Complete a small risk register with likelihood, impact, owner, treatment and residual risk
Backup if inaccessible	C06 — NIST Cybersecurity Framework 2.0
Known limitations	US federal guidance; terminology can be adapted
Classification / age status	Recommended · Classic, technically durable
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

C27 · NIST SP 800-34 Revision 1

Exact title	NIST SP 800-34 Revision 1
Direct URL	https://csrc.nist.gov/pubs/sp/800/34/r1/final
Resource type / domains	Official continuity guidance · Modules J
Topic and subtopic coverage	Evidence-based continuity and disaster recovery planning
Exact path / module / chapter / playlist	Business impact analysis, contingency planning, backup, recovery and testing concepts
Recommended difficulty / estimated time	Intermediate · 4-7 h
Access status	Free
Prerequisites	Asset inventory and risk basics
Why selected	Evidence-based continuity and disaster recovery planning
What to practise after	Write a backup/restore test plan for the capstone and prove one restore
Backup if inaccessible	C06 — NIST Cybersecurity Framework 2.0
Known limitations	US federal guidance; adapt to local context
Classification / age status	Recommended · Classic, technically durable
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W01 · PortSwigger Web Security Academy

Exact title	PortSwigger Web Security Academy
Direct URL	https://portswigger.net/web-security
Resource type / domains	Free interactive academy · Modules F
Topic and subtopic coverage	Best free web-security lab sequence; each concept has a deliberately vulnerable safe lab
Exact path / module / chapter / playlist	All topics and 270+ labs; start with learning paths and Apprentice labs
Recommended difficulty / estimated time	Beginner to advanced · 100+ h available; selected 40-60 h in roadmap
Access status	Free account required
Prerequisites	HTTP, browser DevTools, basic Python
Why selected	Best free web-security lab sequence; each concept has a deliberately vulnerable safe lab
What to practise after	Read topic, solve Apprentice lab without a walkthrough, then write a remediation note
Backup if inaccessible	W02 — PortSwigger Web Security Academy Learning Paths
Known limitations	Labs require a free account and internet; do not target real public sites
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W02 · PortSwigger Web Security Academy Learning Paths

Exact title	PortSwigger Web Security Academy Learning Paths
Direct URL	https://portswigger.net/web-security/learning-paths
Resource type / domains	Structured learning paths · Modules F
Topic and subtopic coverage	Prevents random lab hopping and provides prerequisite-aware order
Exact path / module / chapter / playlist	Curated paths such as Server-side vulnerabilities and SQL injection; choose current paths shown on site
Recommended difficulty / estimated time	Beginner to intermediate · 20-40 h selected
Access status	Free
Prerequisites	W01 basics
Why selected	Prevents random lab hopping and provides prerequisite-aware order
What to practise after	Complete the first two assigned paths in Month 6 and log each lab's root cause and fix
Backup if inaccessible	W01 — PortSwigger Web Security Academy
Known limitations	Path names and lab counts can change
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W03 · PortSwigger All Labs index

Exact title	PortSwigger All Labs index
Direct URL	https://portswigger.net/web-security/all-labs
Resource type / domains	Lab index · Modules F
Topic and subtopic coverage	Direct lab navigation for assessments and retesting
Exact path / module / chapter / playlist	All Apprentice, Practitioner and Expert labs by topic
Recommended difficulty / estimated time	Intermediate · Reference use
Access status	Free
Prerequisites	W01 account
Why selected	Direct lab navigation for assessments and retesting
What to practise after	Select only the exact lab names listed in the daily plan; stop if a lab requires unlearned prerequisites
Backup if inaccessible	W02 — PortSwigger Web Security Academy Learning Paths
Known limitations	Expert labs can be frustrating early; postpone them
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W04 · MDN HTTP documentation

Exact title	MDN HTTP documentation
Direct URL	https://developer.mozilla.org/en-US/docs/Web/HTTP
Resource type / domains	Official documentation · Modules F
Topic and subtopic coverage	Authoritative browser behavior and security-header reference
Exact path / module / chapter / playlist	Methods, status codes, headers, cookies, CORS, CSP, authentication, caching, redirects and connection management
Recommended difficulty / estimated time	Beginner to intermediate · Reference use
Access status	Free
Prerequisites	Basic web development
Why selected	Authoritative browser behavior and security-header reference
What to practise after	Inspect each response header in a local app and explain its security effect
Backup if inaccessible	F10 — MDN HTTP Overview
Known limitations	Large documentation set; use site search and assigned pages
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W05 · Burp Suite documentation

Exact title	Burp Suite documentation
Direct URL	https://portswigger.net/burp/documentation
Resource type / domains	Official tool documentation · Modules F,G
Topic and subtopic coverage	Primary documentation for the main web-security lab tool
Exact path / module / chapter / playlist	Proxy, target/scope, Repeater, Intruder, Scanner, browser, extensions and TLS configuration
Recommended difficulty / estimated time	Beginner to intermediate · 6-10 h selected
Access status	Community edition free; Pro paid
Prerequisites	HTTP and browser proxy basics
Why selected	Primary documentation for the main web-security lab tool
What to practise after	Configure scope for only localhost and PortSwigger lab domains; use Repeater to compare requests
Backup if inaccessible	W01 — PortSwigger Web Security Academy
Known limitations	Some features require paid Pro edition
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W06 · OWASP Web Security Testing Guide

Exact title	OWASP Web Security Testing Guide
Direct URL	https://owasp.org/www-project-web-security-testing-guide/
Resource type / domains	Testing methodology · Modules F,G
Topic and subtopic coverage	Structured professional test methodology emphasizing evidence and scope
Exact path / module / chapter / playlist	Information gathering, configuration, identity, authentication, authorization, session, input validation, error handling, cryptography, business logic and client-side testing
Recommended difficulty / estimated time	Intermediate · Reference use
Access status	Free
Prerequisites	W01
Why selected	Structured professional test methodology emphasizing evidence and scope
What to practise after	Use the relevant WSTG section as a checklist for the local assessment project
Backup if inaccessible	W01 — PortSwigger Web Security Academy
Known limitations	Guide is not a step-by-step beginner tutorial
Classification / age status	Mandatory · Community-maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W07 · OWASP ZAP documentation

Exact title	OWASP ZAP documentation
Direct URL	https://www.zaproxy.org/docs/
Resource type / domains	Official tool docs · Modules F,G
Topic and subtopic coverage	Free alternative to Burp Pro for automated scanning and local security testing
Exact path / module / chapter / playlist	Getting started, proxy, spider, active scan, passive scan, authentication, API and automation
Recommended difficulty / estimated time	Beginner to intermediate · 4-8 h
Access status	Free/open-source
Prerequisites	HTTP and proxy fundamentals
Why selected	Free alternative to Burp Pro for automated scanning and local security testing
What to practise after	Run a passive scan against a local Juice Shop container only and manually validate one finding
Backup if inaccessible	W05 — Burp Suite documentation
Known limitations	Automated scans do not prove exploitability or absence of bugs
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W08 · OWASP Juice Shop

Exact title	OWASP Juice Shop
Direct URL	https://owasp.org/www-project-juice-shop/
Resource type / domains	Vulnerable-by-design app · Modules F,G
Topic and subtopic coverage	Provides a realistic application to test and then remediate through configuration/code review
Exact path / module / chapter / playlist	OWASP Top 10 challenges with hints, score board and deployment options
Recommended difficulty / estimated time	Intermediate · 10-20 h
Access status	Free/open-source
Prerequisites	Docker or Node.js; isolated lab
Why selected	Provides a realistic application to test and then remediate through configuration/code review
What to practise after	Deploy only on localhost or isolated VM; complete five challenges and write defensive fixes
Backup if inaccessible	W09 — OWASP Juice Shop GitHub repository
Known limitations	Never expose intentionally vulnerable software to the public internet
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W09 · OWASP Juice Shop GitHub repository

Exact title	OWASP Juice Shop GitHub repository
Direct URL	https://github.com/juice-shop/juice-shop
Resource type / domains	Source code / setup · Modules F,G
Topic and subtopic coverage	Primary installation and source repository; safer than random third-party copies
Exact path / module / chapter / playlist	README, Docker deployment, configuration, challenge tracking and source code
Recommended difficulty / estimated time	Intermediate · Setup 1-2 h
Access status	Free/open-source
Prerequisites	Docker basics
Why selected	Primary installation and source repository; safer than random third-party copies
What to practise after	Clone an official release/tag, verify checksums where available, run with loopback binding only
Backup if inaccessible	W08 — OWASP Juice Shop
Known limitations	Development branch may change; use official tags or documented release
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W10 · DVWA GitHub repository

Exact title	DVWA GitHub repository
Direct URL	https://github.com/digininja/DVWA
Resource type / domains	Vulnerable-by-design app · Modules F,G
Topic and subtopic coverage	Simple local environment for comparing low/medium/high security levels
Exact path / module / chapter / playlist	Docker/local setup, security levels, SQLi, XSS, CSRF, file inclusion and upload challenges
Recommended difficulty / estimated time	Beginner to intermediate · 5-12 h
Access status	Free/open-source
Prerequisites	Docker or local web stack
Why selected	Simple local environment for comparing low/medium/high security levels
What to practise after	Deploy in host-only VM, complete two vulnerabilities and inspect the corresponding source code
Backup if inaccessible	W09 — OWASP Juice Shop GitHub repository
Known limitations	Outdated interface and some PHP dependencies; do not expose publicly
Classification / age status	Optional enrichment · Old but gold
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W11 · OWASP WebGoat

Exact title	OWASP WebGoat
Direct URL	https://owasp.org/www-project-webgoat/
Resource type / domains	Vulnerable-by-design lessons · Modules F,G
Topic and subtopic coverage	Lesson-oriented alternative when PortSwigger's exploit-first labs feel too abrupt
Exact path / module / chapter / playlist	Lessons on HTTP basics, injection, authentication, access control, CSRF, SSRF and cryptography
Recommended difficulty / estimated time	Beginner to intermediate · 8-15 h
Access status	Free/open-source
Prerequisites	Java or Docker; isolated lab
Why selected	Lesson-oriented alternative when PortSwigger's exploit-first labs feel too abrupt
What to practise after	Complete selected lessons and explain the vulnerable code path and fix
Backup if inaccessible	W01 — PortSwigger Web Security Academy
Known limitations	Requires Java/container resources; use selectively
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W12 · OWASP Dependency-Check

Exact title	OWASP Dependency-Check
Direct URL	https://owasp.org/www-project-dependency-check/
Resource type / domains	Open-source SCA tool · Modules F,I
Topic and subtopic coverage	Introduces supply-chain risk through a real scanner rather than theory
Exact path / module / chapter / playlist	Dependency scanning, NVD data, reports and CI integration concepts
Recommended difficulty / estimated time	Intermediate · 3-6 h
Access status	Free/open-source
Prerequisites	Build system basics
Why selected	Introduces supply-chain risk through a real scanner rather than theory
What to practise after	Scan a small synthetic project with a known-vulnerable dependency in a sandboxed environment, then triage false positives
Backup if inaccessible	W15 — OpenSSF Scorecard
Known limitations	NVD API rate limits and database bootstrap can be slow; use offline sample report if blocked
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W13 · GitHub code security documentation

Exact title	GitHub code security documentation
Direct URL	https://docs.github.com/en/code-security
Resource type / domains	Official documentation · Modules F,I
Topic and subtopic coverage	Directly relevant to publishing a professional portfolio
Exact path / module / chapter / playlist	Dependabot, secret scanning, code scanning, security advisories, supply-chain security and repository hardening
Recommended difficulty / estimated time	Beginner to intermediate · 5-10 h
Access status	Free features vary by repository visibility and plan
Prerequisites	Git basics
Why selected	Directly relevant to publishing a professional portfolio
What to practise after	Enable available free protections on a public sanitized repo and document residual risks
Backup if inaccessible	W14 — Semgrep documentation
Known limitations	Feature availability differs between public/private/enterprise plans
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W14 · Semgrep documentation

Exact title	Semgrep documentation
Direct URL	https://semgrep.dev/docs/
Resource type / domains	Official docs / open-source engine · Modules F,I
Topic and subtopic coverage	Teaches how static analysis rules are written, not just how to run a scanner
Exact path / module / chapter / playlist	Rule syntax, local scans, CI, secrets, supply chain and custom rules
Recommended difficulty / estimated time	Intermediate · 5-10 h
Access status	Open-source engine free; some platform features paid
Prerequisites	Programming and regex
Why selected	Teaches how static analysis rules are written, not just how to run a scanner
What to practise after	Write one simple custom rule for a deliberately insecure local test file and document false positives
Backup if inaccessible	W13 — GitHub code security documentation
Known limitations	Cloud platform and engine features differ
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W15 · OpenSSF Scorecard

Exact title	OpenSSF Scorecard
Direct URL	https://securityscorecards.dev/
Resource type / domains	Open-source project / checks · Modules F,I
Topic and subtopic coverage	Concrete, automated supply-chain hygiene checks for repositories
Exact path / module / chapter / playlist	Checks for branch protection, pinned dependencies, signed releases, tokens, CI and vulnerability disclosure
Recommended difficulty / estimated time	Intermediate · 2-5 h
Access status	Free/open-source
Prerequisites	Git and CI basics
Why selected	Concrete, automated supply-chain hygiene checks for repositories
What to practise after	Run scorecard against a public non-sensitive repository and improve two failing checks
Backup if inaccessible	W13 — GitHub code security documentation
Known limitations	Not all checks apply to small student projects
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W16 · SLSA specification

Exact title	SLSA specification
Direct URL	https://slsa.dev/
Resource type / domains	Supply-chain framework · Modules F,I
Topic and subtopic coverage	Primary framework for reasoning about build provenance and software supply-chain integrity
Exact path / module / chapter / playlist	Build levels, provenance, source, build and dependency threats
Recommended difficulty / estimated time	Advanced · 3-6 h
Access status	Free
Prerequisites	CI/CD and software delivery
Why selected	Primary framework for reasoning about build provenance and software supply-chain integrity
What to practise after	Map one GitHub Actions workflow to SLSA threats and identify missing controls
Backup if inaccessible	W13 — GitHub code security documentation
Known limitations	Specification evolves; use current level documentation
Classification / age status	Advanced reference · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

W17 · OWASP Cheat Sheet Series

Exact title	OWASP Cheat Sheet Series
Direct URL	https://cheatsheetseries.owasp.org/
Resource type / domains	Official checklist library · Modules F
Topic and subtopic coverage	Short, authoritative remediation guidance that pairs well with PortSwigger offense labs
Exact path / module / chapter / playlist	Authentication, session, access control, XSS, SQL injection, CSRF, SSRF, file upload, secrets, logging, TLS and API security cheat sheets
Recommended difficulty / estimated time	Beginner to advanced · Reference use
Access status	Free
Prerequisites	Web security fundamentals
Why selected	Short, authoritative remediation guidance that pairs well with PortSwigger offense labs
What to practise after	After each lab, read the matching cheat sheet and implement one control in the secure app
Backup if inaccessible	C15 — OWASP Application Security Verification Standard
Known limitations	Cheat sheets can be dense; use them after hands-on practice
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

E01 · NIST SP 800-115

Exact title	NIST SP 800-115
Direct URL	https://csrc.nist.gov/pubs/sp/800/115/final
Resource type / domains	Official testing guidance · Modules G
Topic and subtopic coverage	Authoritative methodology for safe security testing and evidence handling
Exact path / module / chapter / playlist	Planning, discovery, vulnerability scanning, validation, reporting and post-test activities
Recommended difficulty / estimated time	Intermediate · 6-10 h
Access status	Free
Prerequisites	Networking and written authorization
Why selected	Authoritative methodology for safe security testing and evidence handling
What to practise after	Create a one-page rules-of-engagement document for your isolated lab before any scanning
Backup if inaccessible	W06 — OWASP Web Security Testing Guide
Known limitations	Published 2008; methodology remains valid but tool examples are outdated
Classification / age status	Mandatory · Old but gold, update tool commands
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

E02 · TryHackMe Pre Security learning path

Exact title	TryHackMe Pre Security learning path
Direct URL	https://tryhackme.com/path/outline/presecurity
Resource type / domains	Interactive path · Modules A,B,E,G
Topic and subtopic coverage	Browser-based guided labs for low-end hardware and early confidence
Exact path / module / chapter / playlist	Computer basics, networking, web fundamentals, Linux and Windows basics, introductory offensive/defensive rooms
Recommended difficulty / estimated time	Beginner · 30-40 h
Access status	Freemium; introductory rooms free, full access may require subscription
Prerequisites	None
Why selected	Browser-based guided labs for low-end hardware and early confidence
What to practise after	Complete only assigned free rooms; use local backups for locked material
Backup if inaccessible	L01 — Linux Journey
Known limitations	Free-room availability and path structure changed in 2026; verify before buying
Classification / age status	Optional enrichment · Revamped 2026
Verification status / date	Rate-limited automated check (HTTP 429); official URL; retry in browser or use backup · checked 2026-10-27

E03 · TryHackMe Cyber Security 101 path

Exact title	TryHackMe Cyber Security 101 path
Direct URL	https://tryhackme.com/path/outline/cybersecurity101
Resource type / domains	Interactive path · Modules E,G
Topic and subtopic coverage	Structured bridge from basics to junior analyst topics; useful only when free rooms are available
Exact path / module / chapter / playlist	Networks, cryptography, Windows/AD, Linux, web, defensive tools and junior roles
Recommended difficulty / estimated time	Beginner · 60-80 h
Access status	Freemium; many rooms subscription-only
Prerequisites	Pre Security foundations
Why selected	Structured bridge from basics to junior analyst topics; useful only when free rooms are available
What to practise after	Use selected free intro rooms as extra practice; do not let subscription replace local projects
Backup if inaccessible	E01 — NIST SP 800-115
Known limitations	Path content and free/paid status can change
Classification / age status	Optional enrichment · Current 2026 path
Verification status / date	Rate-limited automated check (HTTP 429); official URL; retry in browser or use backup · checked 2026-10-27

E04 · Hack The Box Academy

Exact title	Hack The Box Academy
Direct URL	https://academy.hackthebox.com/
Resource type / domains	Interactive learning platform · Modules G,K
Topic and subtopic coverage	High-quality guided modules with downloadable resources and hands-on practice
Exact path / module / chapter / playlist	Introductory modules on networking, Linux, Windows, web, privilege escalation and security fundamentals
Recommended difficulty / estimated time	Beginner to advanced · Varies
Access status	Freemium; some modules free, others require cubes/subscription
Prerequisites	Basic Linux and networking
Why selected	High-quality guided modules with downloadable resources and hands-on practice
What to practise after	Filter for free modules; record local commands and concepts rather than collecting badges
Backup if inaccessible	N09 — Nmap Reference Guide
Known limitations	Pricing and free-module availability vary
Classification / age status	Optional enrichment · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

E05 · HackerOne Disclosure Guidelines

Exact title	HackerOne Disclosure Guidelines
Direct URL	https://www.hackerone.com/disclosure-guidelines
Resource type / domains	Responsible disclosure guidance · Modules G,J
Topic and subtopic coverage	Useful practical introduction to professional disclosure expectations
Exact path / module / chapter / playlist	Safe harbor, authorized testing, reporting expectations and coordinated disclosure concepts
Recommended difficulty / estimated time	Beginner to intermediate · 1-2 h
Access status	Free
Prerequisites	Ethics and scope basics
Why selected	Useful practical introduction to professional disclosure expectations
What to practise after	Draft a disclosure report for a synthetic finding; verify any real program's policy before testing
Backup if inaccessible	E06 — FIRST CVSS specification
Known limitations	Platform policy is not universal law and does not grant authorization for arbitrary targets
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

E06 · FIRST CVSS specification

Exact title	FIRST CVSS specification
Direct URL	https://www.first.org/cvss/
Resource type / domains	Scoring standard · Modules G,J
Topic and subtopic coverage	Standardized severity language; necessary but insufficient for risk prioritization
Exact path / module / chapter / playlist	Base, temporal and environmental metrics; calculator and examples
Recommended difficulty / estimated time	Intermediate · 2-4 h
Access status	Free
Prerequisites	Vulnerability basics
Why selected	Standardized severity language; necessary but insufficient for risk prioritization
What to practise after	Score one synthetic vulnerability, then explain why business context can override the numeric score
Backup if inaccessible	E08 — NVD
Known limitations	CVSS is not a full risk assessment
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

E07 · CVE Program

Exact title	CVE Program
Direct URL	https://www.cve.org/
Resource type / domains	Vulnerability identifier program · Modules G,J
Topic and subtopic coverage	Primary source for identifier process and terminology
Exact path / module / chapter / playlist	CVE IDs, CNAs, records and program information
Recommended difficulty / estimated time	Beginner · 1-2 h
Access status	Free
Prerequisites	Vulnerability concepts
Why selected	Primary source for identifier process and terminology
What to practise after	Take one synthetic finding and explain why it would or would not deserve a CVE
Backup if inaccessible	E08 — NVD
Known limitations	CVE assignment itself does not prove exploitability or priority
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

E08 · NVD

Exact title	NVD
Direct URL	https://nvd.nist.gov/
Resource type / domains	Vulnerability database · Modules G,J
Topic and subtopic coverage	Authoritative public vulnerability-data source for triage exercises
Exact path / module / chapter / playlist	CVE records, CVSS, CPE, references and search APIs
Recommended difficulty / estimated time	Intermediate · 2-5 h
Access status	Free
Prerequisites	CVE/CVSS basics
Why selected	Authoritative public vulnerability-data source for triage exercises
What to practise after	Compare an NVD record with the vendor advisory and note data freshness limitations
Backup if inaccessible	N09 — Nmap Reference Guide
Known limitations	NVD enrichment can lag; API rate limits apply
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

E09 · picoCTF

Exact title	picoCTF
Direct URL	https://picoctf.org/
Resource type / domains	CTF platform · Modules D,G,K
Topic and subtopic coverage	Carnegie Mellon-affordable CTF with approachable early challenges
Exact path / module / chapter / playlist	Beginner cryptography, web exploitation, reverse engineering, forensics and general skills
Recommended difficulty / estimated time	Beginner to advanced · Varies
Access status	Free competition/picoGym account
Prerequisites	Programming and Linux basics
Why selected	Carnegie Mellon-affordable CTF with approachable early challenges
What to practise after	Solve five general-skills or forensics challenges and write independent solution notes
Backup if inaccessible	L15 — pwn.college Dojo index
Known limitations	Competition archive availability and scoreboards change
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

E10 · CyLab Security Academy / picoGym

Exact title	CyLab Security Academy / picoGym
Direct URL	https://www.cylabacademy.org/
Resource type / domains	Practice platform · Modules D,G,K
Topic and subtopic coverage	Allows non-competition practice without waiting for the annual contest
Exact path / module / chapter / playlist	Practice challenges from prior picoCTF competitions
Recommended difficulty / estimated time	Beginner to advanced · Varies
Access status	Free account
Prerequisites	CTF basics
Why selected	Allows non-competition practice without waiting for the annual contest
What to practise after	Choose beginner picoGym problems assigned in the daily plan; do not publish current contest flags
Backup if inaccessible	E09 — picoCTF
Known limitations	Practice catalog may change
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B01 · Microsoft Sentinel documentation

Exact title	Microsoft Sentinel documentation
Direct URL	https://learn.microsoft.com/en-us/azure/sentinel/
Resource type / domains	Official documentation · Modules H
Topic and subtopic coverage	Leading SIEM documentation with free learning paths; use simulated data before cloud deployment
Exact path / module / chapter / playlist	Architecture, data connectors, workspaces, analytics rules, incidents, hunting, KQL, SOAR and content hub
Recommended difficulty / estimated time	Intermediate · 15-30 h selected
Access status	Docs free; Azure/Sentinel usage may incur costs
Prerequisites	Logs and cloud basics
Why selected	Leading SIEM documentation with free learning paths; use simulated data before cloud deployment
What to practise after	Use Microsoft free training data or local sample logs; do not enable paid ingestion without budgets
Backup if inaccessible	B02 — Kusto Query Language overview
Known limitations	Product names and UI change; cloud cost risk
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B02 · Kusto Query Language overview

Exact title	Kusto Query Language overview
Direct URL	https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric
Resource type / domains	Official language docs · Modules H
Topic and subtopic coverage	KQL is essential for modern Microsoft SOC and threat-hunting work
Exact path / module / chapter / playlist	KQL operators, tabular expressions, joins, summarize, let, time-series, parsing and best practices
Recommended difficulty / estimated time	Beginner to intermediate · 10-20 h
Access status	Free
Prerequisites	SQL/log analysis basics
Why selected	KQL is essential for modern Microsoft SOC and threat-hunting work
What to practise after	Write five queries against synthetic security event tables and explain each operator
Backup if inaccessible	B03 — SigmaHQ
Known limitations	Syntax and service-specific tables vary by product
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B03 · SigmaHQ

Exact title	SigmaHQ
Direct URL	https://sigmahq.io/
Resource type / domains	Open-source detection rules · Modules H,K
Topic and subtopic coverage	Vendor-neutral detection engineering with readable, reviewable rules
Exact path / module / chapter / playlist	Generic SIEM detection rule format, rule repository and tooling links
Recommended difficulty / estimated time	Intermediate · 5-10 h
Access status	Free/open-source
Prerequisites	Logs and ATT&CK basics
Why selected	Vendor-neutral detection engineering with readable, reviewable rules
What to practise after	Convert one Sigma rule to a local query and test it against synthetic true/false positive events
Backup if inaccessible	B18 — CISA Known Exploited Vulnerabilities Catalog
Known limitations	Rule quality varies; always validate in your own data
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B04 · Sigma rule creation guide

Exact title	Sigma rule creation guide
Direct URL	https://sigmahq.io/docs/guide/getting-started.html
Resource type / domains	Official rule reference · Modules H,K
Topic and subtopic coverage	Exact syntax reference for writing and reviewing detection rules
Exact path / module / chapter / playlist	Title, ID, status, description, author, date, logsource, detection, condition, falsepositives, level and tags
Recommended difficulty / estimated time	Intermediate · 2-4 h
Access status	Free
Prerequisites	Sigma basics
Why selected	Exact syntax reference for writing and reviewing detection rules
What to practise after	Author one original Sigma rule for a benign lab event and include test data
Backup if inaccessible	B03 — SigmaHQ
Known limitations	Documentation URL or schema can change; use site navigation
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B05 · SBousseaden EVTX-ATTACK-SAMPLES

Exact title	SBousseaden EVTX-ATTACK-SAMPLES
Direct URL	https://github.com/sbousseaden/EVTX-ATTACK-SAMPLES
Resource type / domains	Synthetic/educational log samples · Modules H,K
Topic and subtopic coverage	Rare, credible public event-log dataset for offline detection practice without cloud costs
Exact path / module / chapter / playlist	Windows EVTX samples organized by ATT&CK technique, useful for detection testing
Recommended difficulty / estimated time	Intermediate · 3-8 h
Access status	Free/open-source
Prerequisites	Windows event logs and safe VM
Why selected	Rare, credible public event-log dataset for offline detection practice without cloud costs
What to practise after	Download one sample in an isolated VM, hash it, identify fields, and write a local query
Backup if inaccessible	B02 — Kusto Query Language overview
Known limitations	Third-party repository; verify contents and do not execute unknown binaries; EVTX files are data
Classification / age status	Recommended · Community-maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B06 · CyberDefenders

Exact title	CyberDefenders
Direct URL	https://cyberdefenders.org/
Resource type / domains	Blue-team challenges · Modules H,K
Topic and subtopic coverage	Defensive labs built around realistic evidence and question sets
Exact path / module / chapter / playlist	Digital forensics, incident response, threat hunting, network traffic, endpoint and malware traffic challenges
Recommended difficulty / estimated time	Intermediate to advanced · Varies
Access status	Freemium; some challenges free, some require subscription
Prerequisites	Wireshark, logs and IR basics
Why selected	Defensive labs built around realistic evidence and question sets
What to practise after	Complete one free beginner challenge and write an incident report, not just answers
Backup if inaccessible	N07 — Wireshark SampleCaptures
Known limitations	Free/paid status changes; some datasets are large
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B07 · Blue Team Labs Online

Exact title	Blue Team Labs Online
Direct URL	https://blueteamlabs.online/
Resource type / domains	Blue-team labs · Modules H,K
Topic and subtopic coverage	Short scenario-based labs suitable for weekly assessments
Exact path / module / chapter / playlist	Investigations, phishing, endpoint, SIEM, threat intel and DFIR labs
Recommended difficulty / estimated time	Beginner to advanced · Varies
Access status	Freemium; free and paid labs
Prerequisites	Basic blue-team concepts
Why selected	Short scenario-based labs suitable for weekly assessments
What to practise after	Complete assigned free labs only; substitute local synthetic data if unavailable
Backup if inaccessible	B06 — CyberDefenders
Known limitations	Paid tier/lab catalogue changes; lab platform login required
Classification / age status	Optional enrichment · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B08 · Malware Traffic Analysis

Exact title	Malware Traffic Analysis
Direct URL	https://malware-traffic-analysis.net/
Resource type / domains	Packet capture exercises · Modules B,H,K
Topic and subtopic coverage	Long-running specialist site with realistic traffic-analysis practice
Exact path / module / chapter / playlist	Realistic network captures and malware traffic exercises with questions; use datasets only in isolated analysis
Recommended difficulty / estimated time	Intermediate to advanced · 2-8 h per exercise
Access status	Free
Prerequisites	Wireshark and network fundamentals
Why selected	Long-running specialist site with realistic traffic-analysis practice
What to practise after	Analyze one specified capture in a disposable VM and report IOCs with confidence levels
Backup if inaccessible	N07 — Wireshark SampleCaptures
Known limitations	Contains malware-related artifacts; never execute samples; some links may be old
Classification / age status	Recommended · Old but gold with maintained exercises
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B09 · Zeek documentation

Exact title	Zeek documentation
Direct URL	https://docs.zeek.org/en/master/
Resource type / domains	Official documentation · Modules B,H
Topic and subtopic coverage	Produces structured network telemetry and teaches protocol-aware logging
Exact path / module / chapter / playlist	Architecture, logs, scripting, frameworks, signatures, packet analysis and cluster use
Recommended difficulty / estimated time	Advanced · 10-20 selected pages
Access status	Free/open-source
Prerequisites	Linux and networking
Why selected	Produces structured network telemetry and teaches protocol-aware logging
What to practise after	Install in a low-resource VM or use sample logs; explain one conn.log field
Backup if inaccessible	B10 — Suricata documentation
Known limitations	Installation and version differences; not needed on low-end hardware
Classification / age status	Advanced reference · Maintained
Verification status / date	Rate-limited automated check (HTTP 429); official URL; retry in browser or use backup · checked 2026-10-27

B10 · Suricata documentation

Exact title	Suricata documentation
Direct URL	https://docs.suricata.io/en/latest/
Resource type / domains	Official documentation · Modules B,H
Topic and subtopic coverage	Free IDS/NSM platform for network detection engineering
Exact path / module / chapter / playlist	Installation, configuration, rules, outputs, performance and Lua; Alert and EVE JSON logs
Recommended difficulty / estimated time	Intermediate · 8-15 h
Access status	Free/open-source
Prerequisites	TCP/IP and Linux
Why selected	Free IDS/NSM platform for network detection engineering
What to practise after	Write one benign local alert rule, generate synthetic traffic and validate EVE JSON
Backup if inaccessible	B03 — SigmaHQ
Known limitations	Rule syntax and version-specific features; use matching docs version
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B11 · Autopsy documentation

Exact title	Autopsy documentation
Direct URL	https://www.autopsy.com/support/
Resource type / domains	Official tool docs · Modules H,K
Topic and subtopic coverage	Accessible entry into digital forensics using disk images the student creates
Exact path / module / chapter / playlist	Case creation, disk images, ingest modules, timeline, keyword search, artifacts and reports
Recommended difficulty / estimated time	Beginner to intermediate · 5-10 h
Access status	Free/open-source
Prerequisites	Filesystems and evidence integrity basics
Why selected	Accessible entry into digital forensics using disk images the student creates
What to practise after	Create a small FAT/exFAT or ext4 test image, add benign files, delete one, and recover evidence
Backup if inaccessible	B12 — SANS SIFT Workstation
Known limitations	Older screenshots may differ from current release; never analyze personal disks without permission
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B12 · SANS SIFT Workstation

Exact title	SANS SIFT Workstation
Direct URL	https://www.sans.org/tools/sift-workstation/
Resource type / domains	Forensic toolkit · Modules H,K
Topic and subtopic coverage	Well-known free forensic environment consolidating many analysis utilities
Exact path / module / chapter / playlist	Free Ubuntu-based DFIR workstation, tools, download and documentation
Recommended difficulty / estimated time	Intermediate to advanced · Setup 2-4 h
Access status	Free
Prerequisites	Linux and DFIR basics
Why selected	Well-known free forensic environment consolidating many analysis utilities
What to practise after	Install only if hardware allows; otherwise use Autopsy plus native tools on a disposable VM
Backup if inaccessible	B11 — Autopsy documentation
Known limitations	Large download and Linux familiarity required; SANS courses themselves are paid
Classification / age status	Optional enrichment · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B13 · MISP Project

Exact title	MISP Project
Direct URL	https://www.misp-project.org/
Resource type / domains	Threat intelligence platform · Modules H,K
Topic and subtopic coverage	Widely used platform for structured threat intelligence and indicator context
Exact path / module / chapter / playlist	Indicators, events, attributes, taxonomies, sharing and API concepts
Recommended difficulty / estimated time	Intermediate to advanced · 5-12 h
Access status	Free/open-source; hosting costs possible
Prerequisites	Logging and networking
Why selected	Widely used platform for structured threat intelligence and indicator context
What to practise after	Use official documentation or a local demo; create a synthetic event and map an indicator to an ATT&CK technique
Backup if inaccessible	C18 — MITRE ATT&CK
Known limitations	Do not upload personal or unauthorized data; local hosting can be heavy
Classification / age status	Optional enrichment · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B14 · Elastic detection rules documentation

Exact title	Elastic detection rules documentation
Direct URL	https://www.elastic.co/guide/en/security/current/detection-engine-overview.html
Resource type / domains	Official docs / open-source rules · Modules H,K
Topic and subtopic coverage	Real detection-rule implementation patterns and open-source rule content
Exact path / module / chapter / playlist	Detection engine, rule types, exceptions, timelines, alerts and prebuilt rule repository
Recommended difficulty / estimated time	Intermediate · 5-10 h
Access status	Free docs/open-source rules; Elastic Stack may have paid features
Prerequisites	SIEM basics
Why selected	Real detection-rule implementation patterns and open-source rule content
What to practise after	Read one prebuilt rule, reproduce its logic locally on synthetic logs
Backup if inaccessible	B03 — SigmaHQ
Known limitations	Product UI and rule schema change by version
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B15 · Microsoft SC-200 certification page

Exact title	Microsoft SC-200 certification page
Direct URL	https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/
Resource type / domains	Official certification page · Modules H,J
Topic and subtopic coverage	Optional credential if you specialize in Microsoft security operations
Exact path / module / chapter / playlist	Security operations, incident response and threat hunting objectives using Microsoft Defender/ Sentinel; annual free renewal assessment
Recommended difficulty / estimated time	Intermediate · 20-40 h prep
Access status	Free learning content; exam price region-based
Prerequisites	SOC fundamentals and KQL
Why selected	Optional credential if you specialize in Microsoft security operations
What to practise after	Use free study guide and practice assessment before considering the paid exam
Backup if inaccessible	B01 — Microsoft Sentinel documentation
Known limitations	Exam objectives were refreshed in 2026; check current study guide
Classification / age status	Optional enrichment · Updated 2026-10-21
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B16 · Blue Team Level 1 official certification

Exact title	Blue Team Level 1 official certification
Direct URL	https://www.centri.org/certifications/blue-team-level-1
Resource type / domains	Paid course/certification · Modules H,J
Topic and subtopic coverage	Practical blue-team credential if you later have a training budget and already finished free projects
Exact path / module / chapter / playlist	Phishing, threat intelligence, DFIR, SIEM and incident response with a practical exam
Recommended difficulty / estimated time	Intermediate · 80-120 h
Access status	Paid; GBP 399 listed on official page as checked 2026-10-27; one free resit included
Prerequisites	Networking, logs and Wireshark
Why selected	Practical blue-team credential if you later have a training budget and already finished free projects
What to practise after	Do not buy in Year 1; recreate equivalent labs locally first
Backup if inaccessible	B06 — CyberDefenders
Known limitations	Price, provider brand and access terms may change; not required for this roadmap
Classification / age status	Paid/alternative · Current 2026
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B17 · CISA cybersecurity advisories

Exact title	CISA cybersecurity advisories
Direct URL	https://www.cisa.gov/news-events/cybersecurity-advisories
Resource type / domains	Official advisories · Modules H,J,K
Topic and subtopic coverage	Primary-source reading habit that teaches how incidents are reported in practice
Exact path / module / chapter / playlist	Current technical alerts, TTPs, IOCs and mitigation guidance
Recommended difficulty / estimated time	Intermediate · Ongoing 15 min/week
Access status	Free
Prerequisites	Threat intelligence basics
Why selected	Primary-source reading habit that teaches how incidents are reported in practice
What to practise after	Select one advisory and map two TTPs to ATT&CK and two mitigations to lab evidence
Backup if inaccessible	C18 — MITRE ATT&CK
Known limitations	US government source; not exhaustive global coverage
Classification / age status	Recommended · Continuously updated
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B18 · CISA Known Exploited Vulnerabilities Catalog

Exact title	CISA Known Exploited Vulnerabilities Catalog
Direct URL	https://www.cisa.gov/known-exploited-vulnerabilities-catalog
Resource type / domains	Official vulnerability catalog · Modules G,J
Topic and subtopic coverage	Evidence-based prioritization data that complements CVSS
Exact path / module / chapter / playlist	KEV entries with CVE, vendor/product, due date, name, description and required action
Recommended difficulty / estimated time	Intermediate · 2-5 h
Access status	Free
Prerequisites	CVE basics
Why selected	Evidence-based prioritization data that complements CVSS
What to practise after	Choose one KEV and write a risk-based patch priority note using local context
Backup if inaccessible	E08 — NVD
Known limitations	US federal remediation deadlines do not automatically bind Pakistani or private organizations
Classification / age status	Recommended · Continuously updated
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

B19 · SC-200 Exam Prep YouTube playlist

Exact title	SC-200 Exam Prep YouTube playlist
Direct URL	https://www.youtube.com/playlist?list=PLahhVEj9XNTcHNtYH-inn7xFJactEM7zz
Resource type / domains	Official Microsoft exam-prep video series · Modules H,J
Topic and subtopic coverage	Structured Microsoft-produced video supplement to official Learn paths
Exact path / module / chapter / playlist	SC-200 exam preparation episodes mapped to current skills measured
Recommended difficulty / estimated time	Intermediate · 4-8 h
Access status	Free
Prerequisites	SOC and Microsoft security basics
Why selected	Structured Microsoft-produced video supplement to official Learn paths
What to practise after	Watch selected episodes during Month 8 and Month 10, then complete the linked Learn module
Backup if inaccessible	B15 — Microsoft SC-200 certification page
Known limitations	Certification content can change; use current study guide first
Classification / age status	Optional enrichment · Current playlist, individual dates vary
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL01 · AWS Skill Builder

Exact title	AWS Skill Builder
Direct URL	https://skillbuilder.aws/
Resource type / domains	Official training platform · Modules I,J
Topic and subtopic coverage	Official AWS training hub with current exam prep and foundational security material
Exact path / module / chapter / playlist	Digital courses, exam prep plans, Builder Labs, Cloud Quest and AWS Jam; free and paid content
Recommended difficulty / estimated time	Beginner to advanced · Varies
Access status	Freemium; many digital courses free, labs/subscription can be paid
Prerequisites	Cloud basics
Why selected	Official AWS training hub with current exam prep and foundational security material
What to practise after	Use only free digital courses in Month 9; avoid paid labs until you set a strict budget
Backup if inaccessible	CL02 — AWS security documentation
Known limitations	Catalog and free/paid status changes
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL02 · AWS security documentation

Exact title	AWS security documentation
Direct URL	https://docs.aws.amazon.com/security/
Resource type / domains	Official documentation · Modules I
Topic and subtopic coverage	Primary source for AWS security configuration and service behavior
Exact path / module / chapter / playlist	Security guidance by service, identity, logging, encryption, incident response and best practices
Recommended difficulty / estimated time	Intermediate · Reference use
Access status	Free
Prerequisites	Cloud fundamentals
Why selected	Primary source for AWS security configuration and service behavior
What to practise after	Look up the exact service used in a daily lab and verify security defaults
Backup if inaccessible	CL03 — AWS Well-Architected Security Pillar
Known limitations	Very broad; use search rather than linear reading
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL03 · AWS Well-Architected Security Pillar

Exact title	AWS Well-Architected Security Pillar
Direct URL	https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html
Resource type / domains	Official architecture guide · Modules I,J
Topic and subtopic coverage	Design-level security guidance with review questions
Exact path / module / chapter / playlist	Security foundations, IAM, detection, infrastructure protection, data protection, incident response and application security
Recommended difficulty / estimated time	Intermediate · 8-15 h
Access status	Free
Prerequisites	Cloud fundamentals
Why selected	Design-level security guidance with review questions
What to practise after	Perform a paper review of a fictional small AWS architecture and document ten risks
Backup if inaccessible	CL02 — AWS security documentation
Known limitations	AWS-specific; other clouds need their own architecture guidance
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL04 · AWS Free Tier

Exact title	AWS Free Tier
Direct URL	https://aws.amazon.com/free/
Resource type / domains	Official program page · Modules I
Topic and subtopic coverage	Critical source for avoiding accidental cloud spending
Exact path / module / chapter / playlist	Free-tier offer types, service limits, expiry and account controls
Recommended difficulty / estimated time	Beginner · 1-2 h
Access status	Program terms; some services free temporarily, others always free
Prerequisites	Cloud account basics
Why selected	Critical source for avoiding accidental cloud spending
What to practise after	Read current offer types and configure billing alerts before any hands-on cloud lab
Backup if inaccessible	CL01 — AWS Skill Builder
Known limitations	Terms change; credits/expiry vary; a credit card may still be required
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL05 · Microsoft Azure security documentation

Exact title	Microsoft Azure security documentation
Direct URL	https://learn.microsoft.com/en-us/azure/security/
Resource type / domains	Official documentation · Modules I
Topic and subtopic coverage	Alternative cloud primary source if Azure fits available opportunities better than AWS
Exact path / module / chapter / playlist	Security architecture, identity, network, data, logging, incident response and compliance
Recommended difficulty / estimated time	Intermediate · Reference use
Access status	Free
Prerequisites	Cloud fundamentals
Why selected	Alternative cloud primary source if Azure fits available opportunities better than AWS
What to practise after	Compare one AWS and one Azure control for identity and logging
Backup if inaccessible	CL02 — AWS security documentation
Known limitations	Product names and services change
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL06 · Docker documentation

Exact title	Docker documentation
Direct URL	https://docs.docker.com/
Resource type / domains	Official documentation · Modules C,I
Topic and subtopic coverage	Authoritative container foundation before Kubernetes security
Exact path / module / chapter / playlist	Get started, engine, images, containers, networks, volumes, Compose, build and security
Recommended difficulty / estimated time	Beginner to intermediate · 15-25 h selected
Access status	Free; Docker Desktop licensing may apply in some organizations
Prerequisites	Linux command line
Why selected	Authoritative container foundation before Kubernetes security
What to practise after	Build a tiny image, run isolated container, inspect layers, networks and volumes
Backup if inaccessible	CL07 — Docker Engine security documentation
Known limitations	Menu changes and Docker Desktop license terms differ from Docker Engine
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL07 · Docker Engine security documentation

Exact title	Docker Engine security documentation
Direct URL	https://docs.docker.com/engine/security/
Resource type / domains	Official security docs · Modules I
Topic and subtopic coverage	Primary source for container threat model and hardening controls
Exact path / module / chapter / playlist	Daemon attack surface, namespaces, capabilities, seccomp, AppArmor, rootless mode, images and secrets
Recommended difficulty / estimated time	Intermediate · 6-12 h
Access status	Free
Prerequisites	Docker basics
Why selected	Primary source for container threat model and hardening controls
What to practise after	Run a container as a non-root user, drop capabilities, inspect seccomp and record before/after metadata
Backup if inaccessible	CL06 — Docker documentation
Known limitations	Host kernel distribution affects defaults
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL08 · Kubernetes official documentation

Exact title	Kubernetes official documentation
Direct URL	https://kubernetes.io/docs/home/
Resource type / domains	Official documentation · Modules I
Topic and subtopic coverage	Primary source for cluster architecture and current API behavior
Exact path / module / chapter / playlist	Concepts, tasks, tutorials, reference, kubectl, workloads, services, storage, configuration and security
Recommended difficulty / estimated time	Intermediate · 20-40 h selected
Access status	Free
Prerequisites	Containers and networking
Why selected	Primary source for cluster architecture and current API behavior
What to practise after	Use kind/minikube locally for basic objects; do not pay for a managed cluster in Year 1
Backup if inaccessible	CL09 — Kubernetes security documentation
Known limitations	Version-specific APIs; use supported version docs
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL09 · Kubernetes security documentation

Exact title	Kubernetes security documentation
Direct URL	https://kubernetes.io/docs/concepts/security/
Resource type / domains	Official documentation · Modules I
Topic and subtopic coverage	Primary security reference after architecture fundamentals
Exact path / module / chapter / playlist	Cloud-native security, pod security, RBAC, network policies, secrets, audit, multi-tenancy and supply chain
Recommended difficulty / estimated time	Intermediate · 8-15 h
Access status	Free
Prerequisites	Kubernetes basics
Why selected	Primary security reference after architecture fundamentals
What to practise after	Create a local RBAC role, service account, NetworkPolicy and Pod Security Admission example
Backup if inaccessible	CL08 — Kubernetes official documentation
Known limitations	Some features require cluster configuration that kind/minikube may not fully emulate
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL10 · OWASP Kubernetes Top 10

Exact title	OWASP Kubernetes Top 10
Direct URL	https://owasp.org/www-project-kubernetes-top-ten/
Resource type / domains	Risk reference · Modules I
Topic and subtopic coverage	Risk-oriented checklist for reviewing local manifests and cluster settings
Exact path / module / chapter / playlist	K01-K10: insecure workloads, supply chain, RBAC, policy enforcement, logging, secrets, network, config, vulnerabilities and components
Recommended difficulty / estimated time	Intermediate · 4-8 h
Access status	Free
Prerequisites	Kubernetes basics
Why selected	Risk-oriented checklist for reviewing local manifests and cluster settings
What to practise after	Map ten findings from a synthetic manifest review to K01-K10 and remediation
Backup if inaccessible	CL09 — Kubernetes security documentation
Known limitations	Checklist, not a deployment tutorial
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL11 · Kubernetes Goat

Exact title	Kubernetes Goat
Direct URL	https://madhuakula.com/kubernetes-goat/
Resource type / domains	Vulnerable-by-design K8s lab · Modules I,K
Topic and subtopic coverage	Hands-on local cluster security practice without cloud spending
Exact path / module / chapter / playlist	Scenarios on insecure RBAC, container escape, secrets, supply chain, network policies and observability
Recommended difficulty / estimated time	Advanced · 8-20 h
Access status	Free/open-source
Prerequisites	Kubernetes and Docker basics
Why selected	Hands-on local cluster security practice without cloud spending
What to practise after	Run only in a disposable local cluster; complete two beginner scenarios and remediation notes
Backup if inaccessible	CL10 — OWASP Kubernetes Top 10
Known limitations	Can consume significant RAM; skip if hardware is below 16 GB
Classification / age status	Optional enrichment · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL12 · Trivy documentation

Exact title	Trivy documentation
Direct URL	https://trivy.dev/docs/
Resource type / domains	Official scanner docs · Modules I
Topic and subtopic coverage	Single free scanner for images, dependencies and IaC misconfigurations
Exact path / module / chapter / playlist	Container image, filesystem, repository, IaC, Kubernetes and SBOM scanning; installation and CI examples
Recommended difficulty / estimated time	Beginner to intermediate · 3-8 h
Access status	Free/open-source
Prerequisites	Docker basics
Why selected	Single free scanner for images, dependencies and IaC misconfigurations
What to practise after	Scan a local Dockerfile/image, triage one finding, rebuild and rescan
Backup if inaccessible	W12 — OWASP Dependency-Check
Known limitations	Vulnerability DB freshness and false positives; no scanner proves security
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL13 · Checkov documentation

Exact title	Checkov documentation
Direct URL	https://www.checkov.io/1.Welcome/What%20is%20Checkov.html
Resource type / domains	Official IaC scanner docs · Modules I
Topic and subtopic coverage	Shows how to codify preventive infrastructure controls and validate fixes
Exact path / module / chapter / playlist	Terraform, CloudFormation, Kubernetes, Dockerfile and GitHub Actions policy scanning; custom policies and CI
Recommended difficulty / estimated time	Intermediate · 3-8 h
Access status	Free/open-source; Prisma Cloud features paid
Prerequisites	IaC basics
Why selected	Shows how to codify preventive infrastructure controls and validate fixes
What to practise after	Scan a synthetic Terraform/Kubernetes manifest, remediate three findings and rescan
Backup if inaccessible	CL12 — Trivy documentation
Known limitations	Policy coverage varies by framework and version
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL14 · GitHub Actions security hardening

Exact title	GitHub Actions security hardening
Direct URL	https://docs.github.com/en/actions/security-for-github-actions/security-guides/security-hardening-for-github-actions
Resource type / domains	Official guidance · Modules I
Topic and subtopic coverage	Prevents supply-chain mistakes in the very platform used for the portfolio
Exact path / module / chapter / playlist	Secrets, GITHUB_TOKEN permissions, third-party actions, OIDC, dependency pinning, runners and fork safety
Recommended difficulty / estimated time	Intermediate · 4-8 h
Access status	Free
Prerequisites	Git and CI/CD basics
Why selected	Prevents supply-chain mistakes in the very platform used for the portfolio
What to practise after	Harden one workflow: least permissions, pinned action SHAs, no secrets in logs and dependency scanning
Backup if inaccessible	W13 — GitHub code security documentation
Known limitations	Features differ for public/private/enterprise repos
Classification / age status	Mandatory · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL15 · Terraform documentation

Exact title	Terraform documentation
Direct URL	https://developer.hashicorp.com/terraform/docs
Resource type / domains	Official documentation · Modules I
Topic and subtopic coverage	Common IaC platform for reproducible cloud labs and configuration review
Exact path / module / chapter / playlist	Configuration language, providers, resources, state, modules, CLI and policy guidance
Recommended difficulty / estimated time	Intermediate · 8-20 h selected
Access status	Free/open-source; HCP Terraform has paid tiers
Prerequisites	Infrastructure and Git basics
Why selected	Common IaC platform for reproducible cloud labs and configuration review
What to practise after	Create a local-only Terraform plan for a mock resource; do not apply to a paid cloud account
Backup if inaccessible	CL13 — Checkov documentation
Known limitations	State files can contain secrets; never commit them
Classification / age status	Recommended · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL16 · Cloud Security Alliance Cloud Controls Matrix

Exact title	Cloud Security Alliance Cloud Controls Matrix
Direct URL	https://cloudsecurityalliance.org/research/cloud-controls-matrix
Resource type / domains	Framework/reference · Modules I,J
Topic and subtopic coverage	Cloud-specific control catalog for comparing AWS/Azure implementations
Exact path / module / chapter / playlist	Cloud control domains, control IDs, implementation guidance and mappings
Recommended difficulty / estimated time	Intermediate · 5-10 h
Access status	Free registration may be required for some downloads
Prerequisites	Cloud basics and GRC vocabulary
Why selected	Cloud-specific control catalog for comparing AWS/Azure implementations
What to practise after	Select five CCM controls and map them to AWS/Azure documentation
Backup if inaccessible	CL03 — AWS Well-Architected Security Pillar
Known limitations	Registration and version changes; not law
Classification / age status	Optional enrichment · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL17 · AWS Certified Security - Specialty official page

Exact title	AWS Certified Security - Specialty official page
Direct URL	https://aws.amazon.com/certification/certified-security-specialty/
Resource type / domains	Official certification page · Modules I,J
Topic and subtopic coverage	Advanced exam reference for Year 2+; not a Year 1 target
Exact path / module / chapter / playlist	SCS-C03 exam guide, domains, cost, format, recommended experience and exam prep plan
Recommended difficulty / estimated time	Advanced · 40-80 h prep
Access status	Exam US$300 listed 2026-10-27; training optional/paid
Prerequisites	Five years IT security and two years AWS security recommended
Why selected	Advanced exam reference for Year 2+; not a Year 1 target
What to practise after	Complete free AWS Skill Builder exam prep only after Month 9 foundations and cloud experience
Backup if inaccessible	CL01 — AWS Skill Builder
Known limitations	Region/language and recertification policies change; not entry-level
Classification / age status	Paid/alternative · Current 2026
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

CL18 · AWS recertification page

Exact title	AWS recertification page
Direct URL	https://aws.amazon.com/certification/recertification/
Resource type / domains	Official policy · Modules J
Topic and subtopic coverage	Clarifies renewal obligations before buying an advanced certification
Exact path / module / chapter / playlist	Three-year validity; renewal and maintenance pathways by certification level
Recommended difficulty / estimated time	Reference · 1 h
Access status	Free to read; exam/maintenance may cost
Prerequisites	Certification basics
Why selected	Clarifies renewal obligations before buying an advanced certification
What to practise after	Check current pathway before paying for any AWS exam
Backup if inaccessible	CL17 — AWS Certified Security - Specialty official page
Known limitations	Two official AWS pages describe policy in different detail; verify in your certification account
Classification / age status	Recommended · Updated 2026
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A01 · pwn.college

Exact title	pwn.college
Direct URL	https://pwn.college/
Resource type / domains	Hands-on university platform · Modules A,C,K
Topic and subtopic coverage	University-grade free systems-security curriculum from Arizona State University
Exact path / module / chapter / playlist	Core dojos and current CSE courses; lectures, challenges and browser workspaces
Recommended difficulty / estimated time	Beginner to advanced · Varies
Access status	Free
Prerequisites	Linux, C, Python and assembly depending on dojo
Why selected	University-grade free systems-security curriculum from Arizona State University
What to practise after	Use Computing 101/Linux Luminarium in Year 1; defer Program Security and Software Exploitation to Year 2+
Backup if inaccessible	L15 — pwn.college Dojo index
Known limitations	Course instance names change yearly; use Core Material
Classification / age status	Mandatory for advanced optional track · Maintained 2026
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A02 · pwn.college dojos index

Exact title	pwn.college dojos index
Direct URL	https://pwn.college/dojos
Resource type / domains	Dojo catalog · Modules A,C,K
Topic and subtopic coverage	Best navigation page to pick prerequisites in order
Exact path / module / chapter / playlist	Core Material, Community Material and The Courses list with modules and challenge counts
Recommended difficulty / estimated time	Beginner to advanced · Varies
Access status	Free
Prerequisites	Browser and Linux basics
Why selected	Best navigation page to pick prerequisites in order
What to practise after	Select one module from the assigned dojo and record challenge evidence
Backup if inaccessible	A01 — pwn.college
Known limitations	Dojo catalogue changes; links may require login
Classification / age status	Recommended · Current 2026
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A03 · OpenSecurityTraining2

Exact title	OpenSecurityTraining2
Direct URL	https://ost2.fyi/
Resource type / domains	Free technical training · Modules A,K
Topic and subtopic coverage	University-level low-level security training that is deeper than typical tool tutorials
Exact path / module / chapter / playlist	Architecture, debuggers, reverse engineering, vulnerabilities and exploitation learning paths
Recommended difficulty / estimated time	Intermediate to advanced · 20-100 h
Access status	Free/open-source course materials
Prerequisites	Programming, Linux and architecture
Why selected	University-level low-level security training that is deeper than typical tool tutorials
What to practise after	Use selected Arch1001 videos only after C basics and pwn.college Linux modules
Backup if inaccessible	A01 — pwn.college
Known limitations	Many classes are long and assume C; do not attempt in Month 1
Classification / age status	Advanced reference · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A04 · OST2 Architecture 1001: x86-64 Assembly

Exact title	OST2 Architecture 1001: x86-64 Assembly
Direct URL	https://p.ost2.fyi/courses/course-v1:OpenSecurityTraining2+Arch1001_x86-64_Asm+2021_v1/about
Resource type / domains	Free course · Modules A,K
Topic and subtopic coverage	Best structured x86-64 assembly foundation for later RE, malware analysis and vulnerability research
Exact path / module / chapter / playlist	Registers, stack, functions, control flow, Boolean logic, shifts, repeat instructions, debugging and binary bomb lab
Recommended difficulty / estimated time	Advanced · ~28.5 h
Access status	Free; optional certificate mechanics may vary
Prerequisites	C programming
Why selected	Best structured x86-64 assembly foundation for later RE, malware analysis and vulnerability research
What to practise after	Complete the first four modules and a GDB observation exercise in Year 2+
Backup if inaccessible	A03 — OpenSecurityTraining2
Known limitations	Older recordings but durable architecture concepts; verify tools on current OS
Classification / age status	Advanced reference · Old but gold
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A05 · Malware Unicorn Reverse Engineering 101

Exact title	Malware Unicorn Reverse Engineering 101
Direct URL	https://malwareunicorn.org/workshops/re101.html
Resource type / domains	Free workshop · Modules K
Topic and subtopic coverage	Classic hands-on RE workshop with a strong safety-first lab workflow
Exact path / module / chapter / playlist	Safe VM setup, OS/assembly concepts, malware classes, static/dynamic analysis, disassemblers, debuggers and indicators
Recommended difficulty / estimated time	Advanced · 15-30 h
Access status	Free
Prerequisites	Windows/Linux VMs and assembly basics
Why selected	Classic hands-on RE workshop with a strong safety-first lab workflow
What to practise after	In Year 3+, recreate the lab using current tools and synthetic samples; do not download malware casually
Backup if inaccessible	A04 — OST2 Architecture 1001: x86-64 Assembly
Known limitations	2019 VM images and tools are old; use current official installers and isolated networking
Classification / age status	Advanced reference · Old but gold
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A06 · MIT 6.858 Computer Systems Security OCW

Exact title	MIT 6.858 Computer Systems Security OCW
Direct URL	https://ocw.mit.edu/courses/6-858-computer-systems-security-fall-2014/
Resource type / domains	Graduate university course · Modules E,F,K
Topic and subtopic coverage	Research-paper-driven graduate course that teaches security architecture reasoning
Exact path / module / chapter / playlist	Threat models, OS security, capabilities, information flow, language security, network protocols, web security and research papers
Recommended difficulty / estimated time	Advanced · One semester
Access status	Free
Prerequisites	OS, networking, C and systems programming
Why selected	Research-paper-driven graduate course that teaches security architecture reasoning
What to practise after	Read one assigned lecture note and summarize its threat model, design and evaluation
Backup if inaccessible	A07 — MIT 6.858 official lecture playlist
Known limitations	2014 material; timeless foundations but obsolete tool/browser specifics
Classification / age status	Advanced reference · Old but gold
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A07 · MIT 6.858 official lecture playlist

Exact title	MIT 6.858 official lecture playlist
Direct URL	https://www.youtube.com/playlist?list=PLUl4u3cNGP62K2DjQLRxDNRi0z2IRWnNh
Resource type / domains	YouTube playlist · Modules E,F,K
Topic and subtopic coverage	Direct playlist for the graduate lecture sequence; no invented timestamps
Exact path / module / chapter / playlist	MIT OCW 6.858 Fall 2014 lecture sequence; use exact lecture titles on YouTube
Recommended difficulty / estimated time	Advanced · 20+ lectures
Access status	Free
Prerequisites	Advanced CS fundamentals
Why selected	Direct playlist for the graduate lecture sequence; no invented timestamps
What to practise after	Watch one selected lecture at 1.25x with the matching OCW reading
Backup if inaccessible	A06 — MIT 6.858 Computer Systems Security OCW
Known limitations	2014 publication; update all practical tool details
Classification / age status	Optional enrichment · Old but gold
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A08 · Security Engineering, Third Edition (free online chapters)

Exact title	Security Engineering, Third Edition (free online chapters)
Direct URL	https://www.cl.cam.ac.uk/~rja14/book.html
Resource type / domains	Free book draft · Modules E,J,K
Topic and subtopic coverage	Ross Anderson's systems view connects technical controls to human, economic and organizational failure
Exact path / module / chapter / playlist	Security architecture, psychology, protocols, access control, cryptography, distributed systems, economics, policy and assurance
Recommended difficulty / estimated time	Advanced · Reference/selected chapters
Access status	Free online draft; print edition paid
Prerequisites	Broad technical maturity
Why selected	Ross Anderson's systems view connects technical controls to human, economic and organizational failure
What to practise after	Read one chapter per month in Years 2-3 and write a case analysis
Backup if inaccessible	C06 — NIST Cybersecurity Framework 2.0
Known limitations	Third-edition draft chapters may be updated; not a certification cram book
Classification / age status	Advanced reference · Maintained draft
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A09 · Nightmare binary exploitation course

Exact title	Nightmare binary exploitation course
Direct URL	https://guyinatuxedo.github.io/
Resource type / domains	Free course · Modules K
Topic and subtopic coverage	Large set of progressive exploitation challenges with explanations, useful for true specialization
Exact path / module / chapter / playlist	Linux/Windows exploitation fundamentals, assembly, stack overflows, ROP, heap and challenge walkthroughs
Recommended difficulty / estimated time	Advanced · 30-80 h
Access status	Free
Prerequisites	C, assembly, Linux, GDB and pwn.college prerequisites
Why selected	Large set of progressive exploitation challenges with explanations, useful for true specialization
What to practise after	Attempt only after pwn.college Program Security; use isolated VMs
Backup if inaccessible	A01 — pwn.college
Known limitations	Advanced and sometimes terse; not a Year 1 mastery target
Classification / age status	Advanced reference · Classic
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A10 · RPISEC Modern Binary Exploitation

Exact title	RPISEC Modern Binary Exploitation
Direct URL	https://github.com/RPISEC/MBE
Resource type / domains	University course materials · Modules K
Topic and subtopic coverage	Detailed university lab archive with downloadable materials
Exact path / module / chapter / playlist	Course syllabus, labs and lecture slides on memory corruption and binary exploitation
Recommended difficulty / estimated time	Advanced · One semester
Access status	Free/open-source
Prerequisites	C, Linux and x86
Why selected	Detailed university lab archive with downloadable materials
What to practise after	Use one lab in Year 2+ with a disposable 32/64-bit VM and compare with current mitigations
Backup if inaccessible	A01 — pwn.college
Known limitations	Older VM images and mitigations; modern mitigations require updated setup
Classification / age status	Advanced reference · Old but gold
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A11 · Google Project Zero blog

Exact title	Google Project Zero blog
Direct URL	https://googleprojectzero.blogspot.com/
Resource type / domains	Technical research blog · Modules K
Topic and subtopic coverage	Examples of rigorous vulnerability research methodology and technical writing
Exact path / module / chapter / playlist	Vulnerability research write-ups, root-cause analysis, exploit primitives and mitigation discussions
Recommended difficulty / estimated time	Advanced · Ongoing reading
Access status	Free
Prerequisites	Systems, memory safety and browser/OS internals
Why selected	Examples of rigorous vulnerability research methodology and technical writing
What to practise after	Read one post, extract hypothesis, method, evidence and limitations
Backup if inaccessible	A12 — USENIX Security conference papers
Known limitations	Advanced; many posts require specialized background
Classification / age status	Advanced reference · Continuously updated
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A12 · USENIX Security conference papers

Exact title	USENIX Security conference papers
Direct URL	https://www.usenix.org/conferences/byname/164
Resource type / domains	Conference proceedings · Modules K
Topic and subtopic coverage	Strong source for current peer-reviewed security research with open proceedings
Exact path / module / chapter / playlist	Peer-reviewed security research papers and talks by year
Recommended difficulty / estimated time	Advanced · Ongoing reading
Access status	Many papers freely available; conference attendance paid
Prerequisites	Research methods and domain background
Why selected	Strong source for current peer-reviewed security research with open proceedings
What to practise after	Choose one paper, reproduce a non-dangerous measurement or analysis on synthetic data
Backup if inaccessible	A11 — Google Project Zero blog
Known limitations	Not every paper includes code; reproduction may be hard
Classification / age status	Advanced reference · Annual
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A13 · arXiv cs.CR recent papers

Exact title	arXiv cs.CR recent papers
Direct URL	https://arxiv.org/list/cs.CR/recent
Resource type / domains	Preprint archive · Modules K
Topic and subtopic coverage	Fast way to follow emerging claims, including AI security and systems security
Exact path / module / chapter / playlist	Cryptography and security preprints; use to discover topics, not as peer-reviewed truth
Recommended difficulty / estimated time	Advanced · Ongoing reading
Access status	Free
Prerequisites	Research literacy
Why selected	Fast way to follow emerging claims, including AI security and systems security
What to practise after	Compare one preprint with a later peer-reviewed version or critique its assumptions
Backup if inaccessible	A12 — USENIX Security conference papers
Known limitations	Preprints are not necessarily peer reviewed; quality varies
Classification / age status	Advanced reference · Continuously updated
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A14 · xairy/linux-kernel-exploitation

Exact title	xairy/linux-kernel-exploitation
Direct URL	https://github.com/xairy/linux-kernel-exploitation
Resource type / domains	Curated research index · Modules K
Topic and subtopic coverage	Highly technical index maintained by a practitioner; useful only after years of foundations
Exact path / module / chapter / playlist	Linux kernel exploitation techniques, CVEs, papers, talks and proof-of-concept references
Recommended difficulty / estimated time	Advanced · Reference use
Access status	Free
Prerequisites	Kernel, C and exploitation fundamentals
Why selected	Highly technical index maintained by a practitioner; useful only after years of foundations
What to practise after	Read one concept page and summarize required prerequisites; do not run code on host
Backup if inaccessible	A01 — pwn.college
Known limitations	Dangerous material if misunderstood; use isolated research VMs only
Classification / age status	Advanced reference · Maintained community index
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A15 · OWASP Top 10 for LLM Applications

Exact title	OWASP Top 10 for LLM Applications
Direct URL	https://genai.owasp.org/llm-top-10/
Resource type / domains	Risk reference · Modules F,K
Topic and subtopic coverage	Current application-security risk taxonomy for AI-enabled systems
Exact path / module / chapter / playlist	Prompt injection, sensitive information disclosure, supply chain, data/model poisoning, improper output handling, excessive agency and related AI application risks
Recommended difficulty / estimated time	Intermediate to advanced · 4-8 h
Access status	Free
Prerequisites	Web/API security and LLM basics
Why selected	Current application-security risk taxonomy for AI-enabled systems
What to practise after	Threat-model a simple local RAG app and propose one control for each applicable risk
Backup if inaccessible	C16 — OWASP API Security Top 10
Known limitations	Versioned document; verify current edition before citing
Classification / age status	Advanced reference · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A16 · NIST AI Risk Management Framework

Exact title	NIST AI Risk Management Framework
Direct URL	https://www.nist.gov/itl/ai-risk-management-framework
Resource type / domains	Framework · Modules J,K
Topic and subtopic coverage	Authoritative, voluntary framework for AI risk that complements technical appsec controls
Exact path / module / chapter / playlist	Govern, Map, Measure, Manage; trustworthy AI characteristics and supporting resources
Recommended difficulty / estimated time	Intermediate · 6-12 h
Access status	Free
Prerequisites	Risk and system design basics
Why selected	Authoritative, voluntary framework for AI risk that complements technical appsec controls
What to practise after	Add AI-specific risks to a capstone risk register and map them to NIST AI RMF functions
Backup if inaccessible	A15 — OWASP Top 10 for LLM Applications
Known limitations	Not law and not an appsec testing checklist
Classification / age status	Advanced reference · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A17 · Cisco CCST Cybersecurity official exam page

Exact title	Cisco CCST Cybersecurity official exam page
Direct URL	https://www.cisco.com/site/us/en/learn/training-certifications/exams/ccst-cybersecurity.html
Resource type / domains	Official certification page · Modules B,E,J
Topic and subtopic coverage	Lower-cost optional first certification with a bounded syllabus, but less portable technical proof than a project portfolio
Exact path / module / chapter / playlist	Exam 100-160: essential security principles, network security, endpoint security, vulnerability/risk management and incident handling; US$125 and 50 minutes listed 2026-10-27
Recommended difficulty / estimated time	Beginner · 30-60 h prep
Access status	Exam paid US$125 listed; free Cisco training linked
Prerequisites	Networking and security basics
Why selected	Lower-cost optional first certification with a bounded syllabus, but less portable technical proof than a project portfolio
What to practise after	Use free Cisco training first; consider only after Month 5 if an employer or university requires it
Backup if inaccessible	E01 — NIST SP 800-115
Known limitations	Cisco certification names/pathways are changing in 2026; verify current page and 5-year renewal rules
Classification / age status	Optional certification · Maintained
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A18 · CompTIA Security+ V7 and V8 official pages

Exact title	CompTIA Security+ V7 and V8 official pages
Direct URL	https://www.comptia.org/en-us/certifications/security/
Resource type / domains	Official certification page · Modules E,J
Topic and subtopic coverage	Widely recognized foundational exam, useful as a structured knowledge check but not proof of job-ready skill
Exact path / module / chapter / playlist	V7 SY0-701 current until stated retirement; V8 SY0-801 expected 2026-11-17
Recommended difficulty / estimated time	Beginner to intermediate · Exam 90 min; max 90 questions
Access status	Paid exam; V7 US$404 from CompTIA blog dated January 2025; verify current store
Prerequisites	Core security vocabulary
Why selected	Widely recognized foundational exam, useful as a structured knowledge check but not proof of job-ready skill
What to practise after	Use free Professor Messer course first; consider V8 only after the current objectives are published and Month 12 review
Backup if inaccessible	C01 — Professor Messer SY0-701 Security+ Course Index
Known limitations	V7 retirement English 2027-06-11; V8 launch November 2026; do not buy stale practice material
Classification / age status	Optional certification · Checked 2026-10-27
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A19 · ISC2 Certified in Cybersecurity official page

Exact title	ISC2 Certified in Cybersecurity official page
Direct URL	https://www.isc2.org/Certifications/CC
Resource type / domains	Official certification page · Modules E,J
Topic and subtopic coverage	Low-experience, globally recognized entry-level credential; useful if a scholarship/voucher is available
Exact path / module / chapter / playlist	Five exam domains; no work experience required; current exam outline effective 2026-09-01
Recommended difficulty / estimated time	Beginner · 20-40 h prep
Access status	Exam paid; US$199 listed on official pricing page 2026-10-27; occasional free promotions may exist
Prerequisites	None
Why selected	Low-experience, globally recognized entry-level credential; useful if a scholarship/voucher is available
What to practise after	Use free official flashcards and this roadmap's Month 5 gate before paying
Backup if inaccessible	C03 — ISC2 Certified in Cybersecurity Exam Outline
Known limitations	Annual maintenance fee after certification; exact local price/tax at checkout
Classification / age status	Optional certification · Maintained 2026
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A20 · Microsoft SC-900 official certification page

Exact title	Microsoft SC-900 official certification page
Direct URL	https://learn.microsoft.com/en-us/credentials/certifications/security-compliance-and-identity-fundamentals/
Resource type / domains	Official certification page · Modules E,J
Topic and subtopic coverage	Good free learning path even if you never buy the exam; fundamentals certification does not expire per current Microsoft policy
Exact path / module / chapter / playlist	Security, compliance, identity, Microsoft Entra, Microsoft security and compliance solutions; free practice assessment
Recommended difficulty / estimated time	Beginner · 10-20 h prep
Access status	Free Learn training; exam price region-based
Prerequisites	Cloud and identity basics
Why selected	Good free learning path even if you never buy the exam; fundamentals certification does not expire per current Microsoft policy
What to practise after	Finish all four Learn paths before considering the exam; do not assume Microsoft-specific knowledge equals general security skill
Backup if inaccessible	B15 — Microsoft SC-200 certification page
Known limitations	Price at checkout depends on country; event vouchers are not guaranteed
Classification / age status	Optional certification · Updated 2026-10-21
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A21 · OffSec PEN-200 / OSCP+ official page

Exact title	OffSec PEN-200 / OSCP+ official page
Direct URL	https://www.offsec.com/courses/pen-200/
Resource type / domains	Paid advanced course/certification · Modules G,K
Topic and subtopic coverage	Recognized practical penetration-testing credential for a later specialization, not a beginner goal
Exact path / module / chapter / playlist	PEN-200 modules, challenge labs and 24-hour proctored OSCP+ exam; OSCP lifetime and OSCP+ three-year designation
Recommended difficulty / estimated time	Advanced · 321 h content; 90-day course bundle
Access status	Paid; official page lists US$1749 course+exam bundle and US$1699 standalone exam on 2026-10-27
Prerequisites	Strong Linux, Windows, networking, scripting and ethical testing experience
Why selected	Recognized practical penetration-testing credential for a later specialization, not a beginner goal
What to practise after	Defer until Year 2+ and only after independent authorized assessments and report writing
Backup if inaccessible	E01 — NIST SP 800-115
Known limitations	High cost, renewal/maintenance conditions and exam format can change; free alternatives exist
Classification / age status	Paid/alternative · Maintained 2026
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A22 · INE eJPT official certification page

Exact title	INE eJPT official certification page
Direct URL	https://ine.com/security/certifications/ejpt-certification
Resource type / domains	Official certification page · Modules G,K
Topic and subtopic coverage	More affordable hands-on pentesting exam than OSCP; still not necessary in Year 1
Exact path / module / chapter / playlist	Assessment methodologies, host/network penetration testing, auditing and web application testing objectives
Recommended difficulty / estimated time	Beginner to intermediate · 40-100 h prep
Access status	Paid; official page states an INE subscription and exam voucher are needed; checkout lists voucher/ bundle prices
Prerequisites	Networking, Linux, Windows and basic web security
Why selected	More affordable hands-on pentesting exam than OSCP; still not necessary in Year 1
What to practise after	Consider after Month 7 only if you can demonstrate equivalent skills with local projects and have funds
Backup if inaccessible	E01 — NIST SP 800-115
Known limitations	Official pricing is split between subscription and voucher; checkout promotions change; certification valid three years
Classification / age status	Paid/alternative · Maintained 2026
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A23 · KQL Essentials official Azure Data Explorer playlist

Exact title	KQL Essentials official Azure Data Explorer playlist
Direct URL	https://www.youtube.com/playlist?list=PLKHIhZJiQ5AkOMEC54mQQcAkfg4UXT8uK
Resource type / domains	Official YouTube playlist · Modules H
Topic and subtopic coverage	Short official Microsoft KQL videos complement the textual KQL docs
Exact path / module / chapter / playlist	KQL overview and query operators; use exact episode titles shown on YouTube
Recommended difficulty / estimated time	Beginner to intermediate · 3-8 h
Access status	Free
Prerequisites	Logs and SQL-like query concepts
Why selected	Short official Microsoft KQL videos complement the textual KQL docs
What to practise after	Watch one episode, then reproduce each query on synthetic tables
Backup if inaccessible	B02 — Kusto Query Language overview
Known limitations	Individual publication dates vary; services and UI can change
Classification / age status	Recommended · Current playlist
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27

A24 · SANS DFIR playlist

Exact title	SANS DFIR playlist
Direct URL	https://www.youtube.com/playlist?list=PLrY_AbzZGqt9MJiLjyH-zTrypGLs154e4
Resource type / domains	YouTube playlist · Modules H,K
Topic and subtopic coverage	Free specialist lectures from a reputable training organization; good supplement to official docs
Exact path / module / chapter / playlist	Windows and Linux forensic artifacts, collection and investigation concepts
Recommended difficulty / estimated time	Intermediate to advanced · 4-10 h
Access status	Free
Prerequisites	Windows/Linux internals and evidence integrity
Why selected	Free specialist lectures from a reputable training organization; good supplement to official docs
What to practise after	Select one video, list the artifacts discussed, and verify them on a disposable VM
Backup if inaccessible	B11 — Autopsy documentation
Known limitations	Some SANS courses are paid; webcasts may be older; do not confuse a webcast with certification
Classification / age status	Optional enrichment · Older but technically useful
Verification status / date	Verified HTTP 200; endpoint reachable · checked 2026-10-27